CIMS

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Welcome

CIMS[1] is an integrated set of economic, energy, and materials models designed to provide information to decision makers on the likely response of firms and households to policies that influence their technology acquisition and use decisions. Technology is widely defined to include not just equipment but also buildings and major infrastructure such as transit networks. Thus, it is sometimes described as a technology simulation model, which seeks to reflect how people actually behave rather than how they ought to behave. CIMS is able to provide policy makers with information on: (1) the ability of their policies to achieve specific objectives (like greenhouse gas (GHG) emissions reductions), (2) the likely costs of achieving these objectives, and (3) the uncertainties associated with the model's simulations. CIMS is maintained and operated by the Energy and Materials Research Group at the School of Resource and Environmental Management at Simon Fraser University.

CIMS is based on energy flows through a country's economic system. Like its predecessor, the ISTUM family of models, which form the individual sub-components of CIMS, it tracks the flow of energy, beginning with energy services and production processes through to eventual end-use by individual technologies. Unlike the partial equilibrium ISTUM models, which compete technologies against each other to serve pre-specified demands for end-uses, CIMS is nearly a full equilibrium system that allows the practitioner to include price driven energy and goods and services supply and demand equilibrium, energy trade, and some indirect macroeconomic effects, specifically disposable income and savings multiplier effects.

The model's focus on energy flows through technologies makes it ideal for modelling policies intended to affect energy efficiency, greenhouse gas emissions, and air quality. Energy consumption and emissions of all pollutants are technology specific; unless a model operates on a detailed technology basis, as CIMS does, energy consumption and emission estimates can only be approximated by economic activity.

[1]CIMS originally stood for Canadian Integrated Modelling System. The acronym is now used as a proper name, since over its history the model has been applied to several other countries and regions.

Part I

About

1 History and Development

CIMS was built on top of, and partly out of, a body of technology-simulation work that had been accumulating at the Energy and Materials Research Group (EMRG) at Simon Fraser University since the late 1980s. Understanding how CIMS arrived at its current shape helps explain why some of its modelling conventions are the way they are — the strong emphasis on technology-by-technology accounting, the partial-equilibrium sub-models that still organize the code, the discrete-choice competition equations, and the iterative supply–demand feedback loop all carry forward decisions made over the model's now thirty-plus year history.

1.1 ISTUM origins (late 1980s – 1990s)

CIMS's direct ancestor was the **Industrial Sector Technology Use Model**, ISTUM, a "bottom-up" engineering simulation model of industrial energy demand. ISTUM was initially developed by the US Department of Energy and later maintained at EMRG, working closely with the Canadian Industrial Energy End-Use Data Analysis Centre (CIEEDAC) which provided much of the empirical technology and stock data that the model needed. The first published application is the trial of ISTUM-PC to the British Columbia pulp and paper industry by Jaccard and Roop (1990), which demonstrated the central modelling idea that EMRG would carry forward for the next several decades: represent each end-use service explicitly (process heat, mechanical drive, lighting, etc.) and let candidate technologies compete to provide it, tracking the resulting energy demand and capital stock evolution rather than inferring them from sectoral aggregates.

Over the next decade ISTUM was extended in several directions. Nyboer (1997) worked out the simulation framework in greater detail and applied it to additional sub-sectors. By the early 2000s ISTUM had been built out into a family of sector-specific sub-models covering most of the Canadian economy: primary energy supply (natural gas, coal, crude oil), energy processing (refining, electricity), and end-use demand for energy services (commercial/institutional, transportation, residential, iron & steel, pulp & paper, metal smelting, chemicals, mining, industrial minerals, "other manufacturing"). Each ISTUM sub-model was, on its own, a partial-equilibrium technology simulation — given exogenous demand for the relevant service and exogenous energy prices, it simulated what technology stock would emerge over time. The pieces did not yet talk to each other.

1.2 Construction of CIMS (early 2000s)

The hybrid energy–economy model that became CIMS was led by Chris Bataille as the central work of his PhD at SFU under the supervision of Mark Jaccard and John Nyboer (Bataille 2005). The project's animating question was a methodological one: bottom-up models like ISTUM had strong technological realism but treated demand and prices as exogenous; top-down macro-econometric and CGE models had economic completeness but were largely silent on technology. A "hybrid" model — bottom-up in its technology representation, top-down in its market-clearing behaviour — promised to combine the strengths of both (Rivers and Jaccard 2005b). Bataille's dissertation established design criteria for what such a hybrid should do, surveyed the prior attempts (MARKAL, MESSAGE-MACRO, NEMS), and then built the integration framework on top of the existing ISTUM sub-models (Bataille 2005). Other key members of the EMRG team at SFU who contributed to this phase of the model's development included: Jotham Peters, Rose Murphy, Bryn Sadownik, Alison Laurin, and Nic Rivers.

The principal additions to ISTUM that produced CIMS were:

- **Energy supply–demand equilibration**: the previously isolated sub-models were connected via an iterative loop in which demand-side energy requirements drive supply-side production, supply-side cost-of-production drives the energy prices fed back to the demand side, and the system is iterated until energy prices stabilize within a threshold.
- **Goods-and-services demand feedbacks**: Armington-style price elasticities allow domestic demand for traded goods to respond to changes in domestic production costs (relative to imports), and an empirically estimated relationship between industrial value-added and commercial/residential building permits closes the loop between manufacturing activity and floorspace.
- **Behavioural parameters in the competition**: the technology choice within each sub-model was reformulated as a probabilistic competition with a variance parameter v controlling cost-sensitivity, alongside intangible-cost parameters that capture non-financial preferences and perceived risk. This is the equation now described in Chapter 4.
- **Forward-looking and learning behaviour**: declining-capital-cost and declining-intangible-cost functions were added to represent learning-by-doing and the "neighbour effect" of falling perceived risk as market share grows (Mau et al. 2008); the foresight functions (myopic, average, and discounted expectations of future emissions charges) allow expected policy costs to enter today's investment decisions.

The discrete-choice basis for the technology-competition equation was formalized by Rivers and Jaccard (2005b), which sets out the family of choice models the CIMS competition function belongs to and discusses calibration of v against empirically estimated behavioural parameters. The empirical-parameter programme — collecting stated- and revealed-preference data for personal vehicle and home heating decisions and using it to constrain the v values in those technology competitions — was developed in parallel through several papers in the mid-2000s (Bataille et al. 2006; Mau et al. 2008).

1.3 Evolution

CIMS has been re-implemented several times since its initial construction. The first integrated version used APL for the sub-model calculations and feedback dynamics, with Microsoft Excel/Access for data tables. That generation was flexible but had a difficult user interface. A second version layered Visual Basic and a C++ front-end on top of the APL engine — easier to use but more rigid because data flowed through four different programs in four different languages. A third generation, kept APL as the calculation engine but moved the interface and data storage into Java.

The current implementation has continued this trajectory: the model engine and surrounding extensions are organized as a Python project, with model inputs as CSV and text files, and operator interactions through a marimo notebook layer. The shift in implementation languages over time has tracked broader research-software trends — APL → Visual Basic → Java → Python — while the underlying simulation logic (the partial-equilibrium technology competitions, the iterative supply–demand loop, the discrete-choice competition function, the lifecycle-cost formulation) has remained recognizable from one version to the next.

The Python redevelopment of CIMS has been led by Brad Griffin, Jillian Anderson, and Matt Sniatynski, with considerable support provided by Thomas Budd and Emma Starke as part of their respective PhD research.

1.4 Current status

CIMS continues to be maintained at EMRG and is used by a network of researchers and policy clients across Canada (Energy Modelling Hub 2026). It has been registered in the Energy Modelling Hub's open-models inventory and is part of the Open Insights suite of open-source models, where it sits among the country's small set of national-scope hybrid energy-economy models. Recent extensions include the disaggregation of the Atlantic provinces and territories, expanded coverage of low- and high-temperature industrial heat (see Chapter 34), updated demand and price-elasticity calibrations, and continued evolution of the calibration workflow described in Chapter 39.

1.5 Selected applications

A representative — not exhaustive — list of areas where CIMS has been used to inform policy and academic discussion:

- Evaluation of bottom-up cost curves of US abatement at a range of carbon prices (Murphy and Jaccard 2011).
- Comparisons of bottom-up and top-down estimates of marginal abatement cost for various Canadian sectors and regions.
- Analyses of provincial climate plans and the federal carbon-pricing benchmark.
- Behaviourally realistic assessments of the decarbonization potential of innovative technologies.
- Studies of consumer adoption of new vehicle technologies through discrete-choice analysis (Mau et al. 2008; Horne 2003).
- A book-length academic treatment of the energy-economics research programme in Jaccard (2005).

For a deeper academic introduction to the CIMS modelling philosophy and its place in the broader hybrid-modelling literature, the most useful single reference is still Bataille (2005); Rivers and Jaccard (2005b) is the methodological companion piece on the discrete-choice competition, and Jaccard (2005) gives a book-length statement of the energy-economics research programme which produced CIMS.

2 Model Architecture

2.1 Scale and Scope

CIMS covers the entire Canadian economy and can connect to aggregated representations of other regions (e.g., United States, Europe, or Rest of World). Table 2.1 illustrates the sectors and regions currently covered in CIMS.[1]

Table 2.1: Sectors represented by region

Sectors	**BC**	**AB**	**SK**	**MB**	**ON**	**QC**	**AT**[2]
Coal Mining	+	+	+				+
Natural Gas	+	+	+	+	+	+	+
Petroleum Crude	+	+	+	+	+		+
Petroleum Refining	+	+	+		+	+	+
Electricity	+	+	+	+	+	+	+
Ethanol	+	+	+	+	+	+	+
Biodiesel	+	+	+	+	+	+	+
Hydrogen	+	+	+	+	+	+	+
Mining	+		+	+	+	+	+
Industrial Minerals	+	+			+	+	+
Chemical Products	+	+	+		+	+	
Pulp and Paper	+	+			+	+	+
Metal Smelting	+			+	+	+	+
Iron and Steel			+		+	+	
Light Industrial	+	+	+	+	+	+	+
Construction	+	+	+	+	+	+	+
Residential	+	+	+	+	+	+	+
Commercial	+	+	+	+	+	+	+
Transportation Passenger	+	+	+	+	+	+	+
Transportation Freight	+	+	+	+	+	+	+
Waste	+	+	+	+	+	+	+
Agriculture	+	+	+	+	+	+	+
Forestry	+	+	+	+	+	+	+

[1] Canadian territories are not currently represented in CIMS.

TODO: Could include major metro regions like Toronto, Montreal, and Vancouver to represent city building and transportation policies.

While the model is simple in operation, it can appear complex because it is technologically explicit and covers the whole economy. CIMS represents a detailed list of technologies (fridges, light bulbs, cars, industrial motors, steel furnaces, buildings, power plants, etc.), and includes their linkages to each other. The model is especially large for the industrial sectors because industry has the largest variety of technologies.

CIMS highlights the interplay of energy supply and demand because energy-related GHG emissions are currently a key policy concern. Thus, the model focuses on the interaction between sectors that use energy (industrial, residential, commercial / institutional, and transportation sectors) and sectors that produce or transform energy (electricity generation, fossil fuel supply, oil refining, natural gas processing, biofuels, and hydrogen). This interaction is shown in Figure 2.1. A policy that seeks to influence energy supply and demand may also influence the cost of producing goods and services, thus affecting their price and final demand. To reflect these interactions, CIMS includes a goods and services final demand feedback loop, composed of Armington elasticities for traded goods and empirically driven relationships between the traded and non-traded sectors.

2.2 Simulation: Six Steps

As a simulation model, CIMS simulates the evolution of technologies acquired to meet energy services, and it uses both engineering and economic information. The engineering information details the unit energy consumption of each energy service technology and explains, using flow models of technologically heterogeneous sectors, how final energy service demands, intermediate products, and energy supply are related. The economic information determines which technologies are preferable in terms of cost minimization. While CIMS is not an optimization model, it can be adapted to produce results like those of an optimization model by calling on certain built-in functions. These functions are known as winner-take-all functions, and they give all new market shares in a given competition to the technology or process with the least cost.

CIMS is designed to mimic a sector or industry by representing a number of industrial establishments, residential or commercial buildings, and delivery of goods or personal travel within a defined region. The model focuses its analysis on the aggregate energy end-use requirements of that industry or sector. For example, in industry the major end-uses are process heat, motive force, space heat, and lighting. In the pulp and paper sector, the major end-uses are steam heat or mechanical energy to digest wood fibre into pulp, mechanical energy to convey and form pulp into paper, mechanical energy to press moisture out of formed pulp or

[2] Atlantic includes New Brunswick, Nova Scotia, Prince Edward Island, and Newfoundland and Labrador.

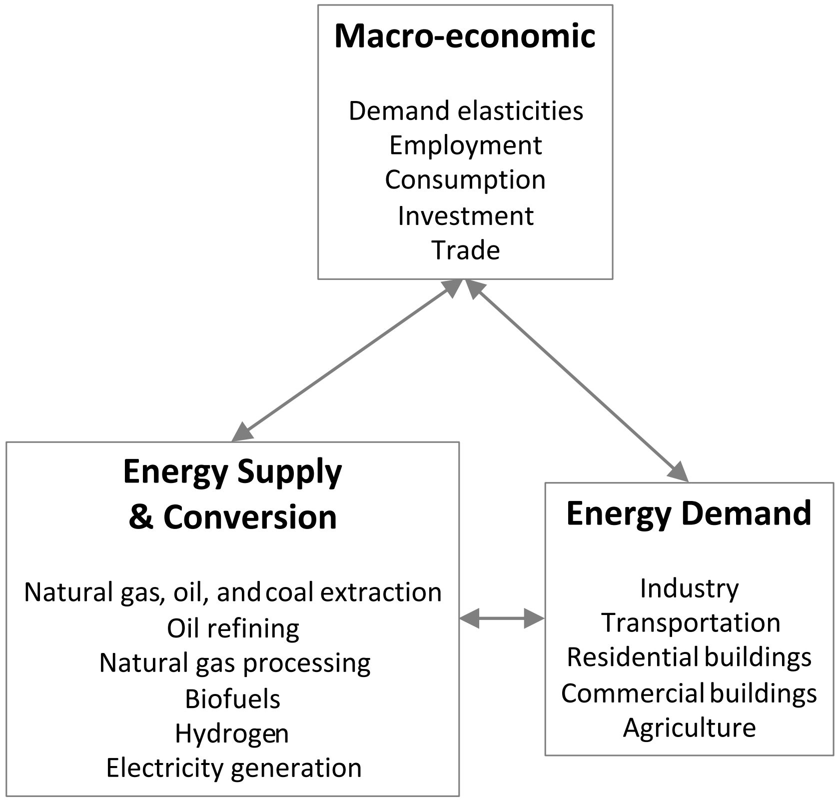

Figure 2.1: General model structure

paper, steam heat to dry pulp or paper, space heat, and lighting. In residential and commercial buildings, the primary end-uses include space heating and cooling, hot water production, refrigeration, and other assorted appliances.

A CIMS simulation comprises six basic steps:

1. The baseline macroeconomic drivers, energy supply drivers, and energy service technology characteristics are set for the desired time frame (15 years, 20 years, etc.) in five-year increments.

    a. Technologies are represented in the model in terms of the quantity of energy service they provide. This could be, for example, vehicle kilometers traveled, tonnes of paper, or square metres of floor space heated and cooled. Then, a forecast of growth in energy service demand is used to drive future demand for energy services, usually in five-year increments.

    b. Macro-economic drivers include income, economic output, structural evolution of sectors, interest rates, employment rates, regulations, etc. They account for all energy service demands in the energy demand module.

c. Energy supply drivers include international commodity prices, domestic market developments, forecasts of costs and availability of marginal energy supplies, and the resulting forecasts of domestic prices.

d. Energy service technology characteristics include the date a technology or process will be available, technology energy efficiencies and capital / operating costs, firm and household time preferences (discount rates) for discretionary and non-discretionary expenditures (new and retrofit), and other behavioural parameters (perceived risks and consumer surpluses of different technologies, interdependence of certain technology choices that serve different energy services, such as domestic water heating and domestic space heating).

2. The energy demand module (industry, commercial, residential, and transportation) is run for the initial driver values (including energy supply prices).

   a. In each future period, a portion of the initial year's stock of technologies is retired. Retirement depends only on age. The remaining technology stocks in each period are subtracted from the forecast energy service demand and this difference determines the amount of new technology stocks in which to invest.

   b. Prospective energy service technologies compete to fill the new service demand. The objective of the model is to simulate this competition so that the outcome approximates what would happen in the real world. Here the model relies heavily on market research of past and prospective firm and household behavior. Technology costs depend on recognized and financial costs, but also on identified differences in non-financial preferences (e.g., differences in the quality of lighting from different light bulbs) and risks (one technology is seen as more likely to fail than another). Even the determination of financial costs is not straightforward as time preferences (discount rates) can differ depending on the decision maker (household vs. firm) and the type of decision (non-discretionary vs. discretionary). Consequently, the competition is simulated probabilistically; new stock purchases are distributed among prospective technologies with financial costs being just one of several factors in determining market share.

   c. In each period, a similar competition occurs with remaining technology stocks to simulate retrofitting (if desirable and likely from the firm or household's perspective). The same financial and non-financial information is required, except that the capital costs of remaining technology stocks are excluded, having been spent earlier when the remaining technology stock was originally acquired.

3. The demand models then send their requirements for energy to the energy supply models. The supply models send back corresponding energy prices. If the changes in energy prices are over a set threshold value for equilibrium (usually 5%), the demand models are then rerun with the new energy prices. This process is repeated until all energy price changes fall under the threshold value. Figure 2.2 describes this process:

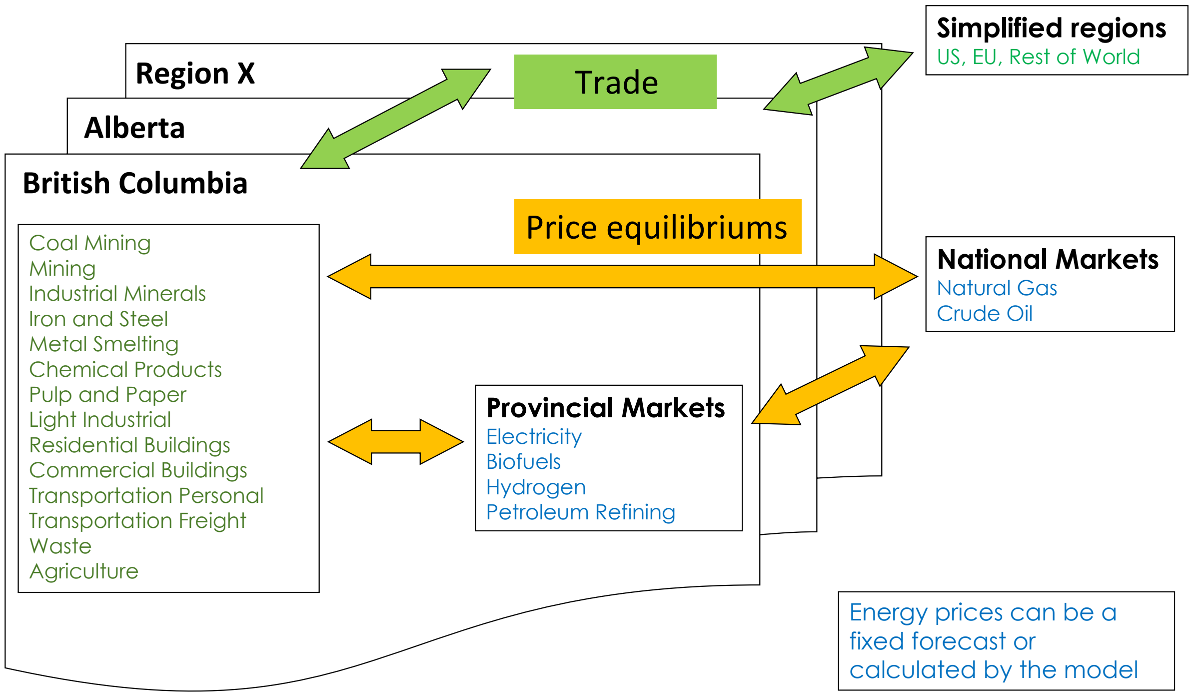

Figure 2.2: Energy Supply and Demand

4. Policy or market driven energy price changes may change the lifecycle cost of some energy services, thereby changing the end-use price for these services. If the change in lifecycle cost from the business-as-usual case is beyond a pre-set threshold price elasticities for these services are applied, thereby changing the demand for these services. This new demand is sent back to the supply and demand models for re-calculation.

5. The model reiterates step 2, 3 and 4 until all variables stabilize. It then proceeds to the next time period. Steps 1 through 5 are repeated for all time periods.

6. Output is provided for new and total capital stock, energy use and trade flows, GHG emissions, levelized lifecycle costs by service and technology, and energy prices.

Part II

Methodology

3 Theory

3.1 CIMS in the energy-economy modelling landscape

CIMS sits in a family of integrated energy-economy models that policy analysts use to study questions like the cost of greenhouse gas abatement, the energy-sector consequences of carbon pricing, or the long-run response of capital stocks to regulation. These models broadly fall into three traditions, and CIMS is built as a deliberate hybrid of them. Locating CIMS within this landscape — and being explicit about what it borrows from each tradition and what it deliberately rejects — is the easiest entry point into the conceptual machinery that the rest of this chapter develops in detail.

3.1.1 Top-down general-equilibrium models

The top-down tradition is dominated by *computable general equilibrium* (CGE) models, which represent the entire economy as a system of interconnected production and consumption functions calibrated to a social accounting matrix. Producers and consumers respond to price changes through elasticities of substitution; markets for goods, labour, and capital clear endogenously; and the model returns a new equilibrium price-quantity vector consistent with the imposed policy shock. CGE models are well suited to questions about how a carbon price reshapes terms of trade, sectoral value-added, household income distribution, or the inter-sectoral reallocation of capital — questions whose answers depend on circulating effects across the whole economy.

What CGE models do not do well is *technology*. An entire production sector is typically represented as a Cobb-Douglas or nested CES production function over a handful of aggregate inputs, and "switching to a cleaner technology" appears, if at all, as movement along an aggregate substitution curve rather than as the retirement of one furnace and the purchase of another. This makes a pure CGE model awkward for analysing policies whose mechanism runs through specific pieces of equipment — vehicle fuel-economy standards, equipment efficiency regulations, capital subsidies for heat pumps, retrofits to industrial boilers. The aggregate elasticities can be made to *mimic* the average response, but the model cannot tell the analyst *which* technologies turned over to produce it.

3.1.2 Bottom-up optimization models

The bottom-up optimization tradition — the MARKAL / TIMES family, MESSAGE, OSeMOSYS, and similar — takes the opposite stance. Each represents the energy system as an explicit graph of technologies with engineering-realistic capital costs, conversion efficiencies, and stock dynamics, then solves a (typically linear) programme that chooses the least-cost mix of technologies to satisfy exogenous service demands subject to constraints. This delivers full technological resolution and a clean, defensible objective function: the answer is the cost-minimising solution under the modeller's assumptions.

The price of that clarity is a strong normative assumption about decision-making. Investment is made by an omniscient, infinitely patient social planner using a single social discount rate; near-identical alternatives are split arbitrarily by the solver rather than smeared across decision-makers; and there is no formal place for non-financial preferences, perceived risk, or the heterogeneity of actual firms and households. Real adoption patterns are observably smoother and more dispersed than the corner solutions that linear-programme formulations tend to produce, and the reasons — uncertainty, information frictions, financing constraints, brand loyalty, perceived reliability — are precisely the factors a least-cost solver does not represent.

3.1.3 Bottom-up simulation models

The third tradition, *bottom-up simulation*, is exemplified by the US Energy Information Administration's National Energy Modeling System (NEMS) and by CIMS's own ancestor, the ISTUM family. These models keep the explicit technology representation of the optimization tradition but drop the social-planner assumption. Technology choice is modelled instead as it is observed to occur: probabilistically, on the basis of decision-maker-specific perceived costs, with the share of new investment going to each option depending on how that option compares to its rivals on both financial and non-financial grounds. There is no claim that the resulting mix is socially optimal — only that it is a plausible representation of what real decision-makers would actually do.

This is the tradition CIMS most directly belongs to.

3.1.4 CIMS as a hybrid

What distinguishes CIMS from a pure bottom-up simulation is that it grafts a partial-equilibrium feedback structure onto the simulation core, producing what its designers called a **hybrid energy-economy model** (Bataille 2005; Rivers and Jaccard 2005b). The bottom-up half is a network of sector-specific simulation sub-models in which candidate technologies compete to provide energy services — process heat, motive force, space conditioning, freight movement, and so on — using a discrete-choice market-share equation whose variance parameter v is calibrated against empirically observed adoption patterns rather than assumed (see Chapter 4). The top-down half is the iterative price-quantity equilibration that connects the sub-models: aggregate energy demand from the demand-side sub-models drives the energy-supply sub-models, the supply sub-models return prices that feed back into the next iteration's lifecycle-cost calculations, and Armington-style elasticities allow demand for traded goods to respond to the relative cost of domestic versus imported production. The system is iterated until energy prices and goods demands stabilise within a tolerance.

This positioning gives CIMS a different mix of strengths and weaknesses from its neighbours. Compared with a CGE model, CIMS sees individual furnaces, vehicles, and building shells, and so can speak directly to technology-specific policies; but it is silent on labour markets, the full inter-industry circulation of value-added, and most of what a CGE practitioner would call "general equilibrium." Compared with a MARKAL-style optimization model, CIMS produces smoother, more behaviourally realistic adoption patterns and can be calibrated against revealed-preference data, but it makes no claim that the resulting technology mix is socially optimal — only that it is what real decision-makers, with their actual heterogeneity and risk perceptions, would plausibly choose.

The two framings are not mutually exclusive in practice. Where the analyst wants an optimization-style answer — the lowest-cost technology mix that satisfies a constraint — CIMS can be coaxed in that direction by setting v very high in the relevant competitions, which collapses the probabilistic share equation toward a winner-take-all rule. This is discussed in its own right below and formalised in Chapter 4. The point for now is that the probabilistic-versus-deterministic framings of technology choice sit on a continuum in CIMS rather than being separate model paradigms, and which end of the continuum is appropriate is a calibration question, not a structural one.

The remainder of this chapter develops the conceptual basis that makes this hybrid structure work: the choice of energy *services* as the unit of analysis, the discrete-choice / Weibit family of competition equations, the iterative price-quantity loop, the distinction between behavioural and financial discount rates, and the role of intangible costs in calibrating non-financial preferences.

3.2 The discrete-choice family of technology competition models

The new-stock competition equation in Chapter 4 is, mathematically, a normalised inverse-power function on lifecycle cost: the market share of each technology is proportional to its lifecycle cost raised to $-v$, divided by the same quantity summed over all competitors. That short description hides a deliberate methodological choice. CIMS is not the only model that converts an array of candidate technology costs into a share split, and the form it uses sits inside a well-developed family of *discrete-choice* models drawn from microeconometrics. This section unpacks where that family comes from, where the CIMS competition equation sits inside it, and why that placement matters when calibrating or interpreting model output.

3.2.1 Random utility theory and the multinomial logit

The discrete-choice literature begins with the observation that an analyst trying to predict which option a decision-maker will pick from a small menu of alternatives never has all the information available to that decision-maker. There are preferences, situations, and constraints that are invisible to the model but visible to the chooser. The standard response, formalised by McFadden (1974), is to model the latent utility of each option as a sum of a deterministic component the analyst can compute — a function of observable attributes like price, efficiency, and capacity — and a stochastic component representing everything the analyst cannot see. The chooser is assumed to pick the option with the highest realised utility; the analyst can then predict only the *probability* that each option is chosen, given the distribution of the unobserved component.

The functional form of the choice probability depends entirely on what distribution one assumes for that unobserved error. If the errors are independent and identically distributed Gumbel (extreme-value type I), a tidy closed-form expression drops out: the probability that option k is chosen equals $e^{V_k}/\sum_j e^{V_j}$, where V_k is the deterministic utility of option k. This is the *multinomial logit* model, and it is the mainstay of applied discrete choice in transportation, marketing, and energy. Its appeal is partly empirical — it fits well in many settings — and partly computational, since the likelihood functions for logit models are globally concave and easy to estimate.

3.2.2 The Weibit and the CIMS power form

The multinomial logit's tractability comes from one specific distributional assumption. Change the assumption and a different choice probability falls out. If the unobserved component enters utility *multiplicatively* rather than additively, and is distributed Weibull rather than Gumbel, the closed-form result is a power expression rather than an exponential one: the probability that option k is chosen equals $V_k^{-v}/\sum_j V_j^{-v}$, where V_k is now a *cost* rather than a utility and $v > 0$ plays the same scale role the scale parameter plays in logit. This is the **Weibit** model, derived in closed form by Castillo et al. (2008), and it is this form that CIMS uses for technology competition.

The two forms are related: taking logs of the cost terms in the Weibit equation produces an expression that is algebraically identical to a multinomial logit on ln V_k with coefficient $-v$. So an equivalent way to describe the CIMS competition is "a multinomial logit on log-lifecycle-cost." The choice between describing it as Weibit-on-cost or logit-on-log-cost is largely a matter of which side of the algebra the analyst finds more natural. The empirical content — that share splits depend on cost *ratios* rather than cost differences, and that the dependence is parameterised by a single scale parameter v — is the same either way.

CIMS adopted the power form partly for historical reasons — the equation was inherited and refined from the ISTUM family of models that predated CIMS — and partly for its behavioural properties. Working in cost-ratio space makes the share split scale-free: doubling all lifecycle costs in a competition (because, say, currency units changed) leaves the split unchanged, which is the behaviour one wants from a competition equation. The same is not true of a logit applied directly to costs in dollar units. The power form does come with one awkwardness: V_k^{-v} is only well-defined for strictly positive V_k (or strictly negative), so when a competition includes technologies whose lifecycle cost is negative — for example because intangible-cost adjustments offset the financial cost — a separate treatment has to be applied. TODO: cross-reference the section where the negative-lifecycle-cost handling is described.

3.2.3 The heterogeneity parameter v

The parameter v deserves attention in its own right, because it is the single most consequential scalar in the technology competition. Algebraically, v is the scale of the underlying error distribution: small v corresponds to large unobserved variance — the analyst's deterministic cost calculation captures relatively little of what the chooser is actually responding to — and large v corresponds to small unobserved variance, with the deterministic cost calculation explaining almost everything.

The two limits are instructive. As $v \to 1$, the share split flattens toward an even split across all available technologies regardless of cost, and the model effectively concludes that lifecycle-cost ratios contain no information about decision-maker behaviour. As $v \to \infty$, the share split sharpens onto the single lowest-cost option, and the competition becomes a winner-take-all rule equivalent in outcome to a linear-programme corner solution. Real-world adoption patterns sit between these extremes, and the practical calibration question is *how far* between — which is exactly what v is asked to encode.

A second, often-missed point about v is that it varies across competitions. The "right" v for a decision over residential furnaces is not the same as the right v for a decision over industrial process-heat technologies, because the underlying decision-makers are different (households versus capital-budgeting committees), the stakes are different, and the set of unobserved factors driving the decision is different. CIMS exposes v as a per-competition parameter so that the model can carry different behavioural assumptions for different decisions, rather than imposing a single global "rationality dial."

3.2.4 Empirical interpretation and calibration

Because v has a microeconomic interpretation as a scale parameter on revealed preferences, it can in principle be estimated from data. The CIMS programme has pursued this most aggressively for personal-vehicle and home-heating decisions, where stated- and revealed-preference surveys are tractable and where the choice sets are small enough to estimate cleanly (Rivers and Jaccard 2005b; Bataille et al. 2006; Mau et al. 2008). For competitions where direct estimation is not feasible — for example, deep inside an industrial process where the "decision-maker" is a capital-allocation committee whose criteria are not surveyable — v is calibrated indirectly against observed historical market shares as part of the broader workflow described in Chapter 39.

3.2.5 Limitations worth flagging

The Weibit equation, like the multinomial logit it is closely related to, inherits the *independence of irrelevant alternatives* (IIA) property: the ratio of choice probabilities between any two options depends only on those two options' costs and is unaffected by what other options are in the choice set. This is convenient algebraically but unrealistic when the choice set contains close substitutes — adding a second "green" option to a menu should mostly steal share from the existing green option, not from the unrelated brown options, yet IIA implies proportional substitution from all incumbents. Nested-logit and mixed-logit variants address this in the broader literature; CIMS does not currently use them, and the practical mitigation is to be careful when structuring the technology hierarchy so that close substitutes are not all thrown into the same competition node.

3.3 Iterative price-quantity equilibration

A model that represents technology choice probabilistically inside individual sub-models still needs a way to make those sub-models behave consistently with one another. The lifecycle cost of an electric arc furnace depends on the price of electricity; the price of electricity depends on how much of it the steel sector and others actually buy; how much they buy depends in turn on which furnace technologies won their respective competitions — which depends on the price of electricity. This is a fixed-point problem, and CIMS solves it by iteration rather than by simultaneous solution. The mechanism deserves its own conceptual treatment because it is what gives CIMS its "partial-equilibrium" character without committing it to the full general-equilibrium machinery of a CGE model.

3.3.1 The fixed-point structure of the problem

At any given simulation year, the system contains two coupled blocks. The *demand-side* sub-models (residential, commercial, transportation, industrial) each take energy prices as exogenous inputs, run their technology competitions, and produce a vector of energy demands (so much electricity, so much natural gas, so much refined petroleum, etc.) as output. The *supply-side* sub-models (electricity generation, refining, natural-gas processing, biofuels, hydrogen) each take energy demands as exogenous inputs, run their own technology competitions to decide how the demand is met, and produce a vector of energy prices as output. Each block, in isolation, is a partial-equilibrium calculation: it treats the other block's outputs as given. Coupling the two creates a circular dependency that the partial-equilibrium calculations do not themselves resolve.

CIMS resolves the circularity by *iteration to a fixed point*. The model uses an initial estimate of the energy-price vector, runs the demand-side at those prices, feeds the resulting demands into the supply-side, observes the new prices the supply block produces, and compares them to the previous input estimate. If the two vectors agree within a tolerance — historically 5% on every energy carrier — the system is declared equilibrated for that year and the simulation advances to the next time step. If they disagree, the new prices replace the old input and the demand-supply pass is repeated. In the latest version of CIMS, this price comparison is not only completed for energy carriers, but for every service lifecycle cost in the model. This better ensures all levels of the model are in equilibrium.

The iteration process is illustrated schematically by the demand-supply feedback figure in Chapter 4, and the same logic governs the goods-demand feedback described in the next paragraph.

The current implementation adds a structural wrinkle to this picture. CIMS now represents the model as a directed graph (using Python's NetworkX package), and each demand-supply pass is realised as a pair of graph traversals — top-down and bottom-up — that require the underlying graph to be acyclic (that is, no loops are present in the graph). Deliberate cycles do, however, exist in the model design: for example, a biodiesel-fuelled tractor consumes biodiesel as an input, and the biodiesel sub-model in turn calls back on tractor service as part of its own production. To run the traversal at all, CIMS applies a small set of loop-resolution heuristics that pick one node in each cycle to evaluate first, using estimated values from its parents on that first pass. Because that estimate is only an approximation of the true equilibrium value, the outer iteration is required to converge several times in succession rather than just once, giving the values inside the broken loop a chance to stabilise alongside the price vector.

3.3.2 Goods and services demand feedbacks

A second product feedback loop sits alongside the energy iteration. When a policy shifts production costs in one of the trade-exposed sectors — pulp and paper, primary metals, chemicals — the price of the domestically produced good moves relative to its imported substitute. Armington-style elasticities then shift the demand for the domestic good, which in turn changes the activity level driving the demand-side technology competitions in the affected sector, which can change energy demand and so on. CIMS treats this loop in the same iterative spirit: the goods-price-vs-quantity relationship is evaluated, the activity vector is updated, and the energy-price iteration restarts under the new activity levels. Convergence in practice depends on having well-behaved elasticities and a finite-amplitude policy shock; the elasticities themselves are documented in the goods-and-services demand feedbacks chapter.

3.3.3 Why use partial-equilibrium rather than general

A full CGE model would close *every* circular dependency in the economy simultaneously — energy prices and demands, but also labour markets, capital allocation, terms of trade, exchange rates, household consumption baskets — typically by feeding a single nonlinear system of equations to a solver and asking for the price-quantity vector that zeros out all the market-clearing residuals at once. CIMS does not do that. It closes the energy market and the trade-in-goods market against the technology sub-models, and treats everything else (labour, capital cost, exchange rates, aggregate macroeconomic activity) as exogenous drivers set at the start of the run. Equilibrium is reached over the markets the model considers endogenous, but not over the whole economy.

The trade-off is the same one that runs through the rest of the chapter. Closing fewer markets keeps the technology sub-models tractable and lets them carry the engineering and behavioural detail that distinguishes CIMS from a CGE model, at the cost of being silent on the macroeconomic feedbacks a CGE model would capture. Closing more markets would push the model toward CGE territory and probably toward greater aggregations of technology, which would decrease the model's utility for evaluating regulatory policies. Further, it is sometimes advantegeous to disconnect a sector or region from the wider economy and explore policy effects in isolation. This flexibility of scenario design and modelling approach is one of the core strengths of CIMS.

3.3.4 Convergence and tolerance

A couple of practical points are worth noting about the iteration itself. First, convergence is not guaranteed in general — a competition equation with high v and tightly coupled supply and demand can in principle exhibit limit cycles, where the share of, say, electric vehicles oscillates between iterations because each side over-corrects to the other. In practice CIMS rarely encounters this regime, partly because the smoothing imposed by the Weibit equation (intermediate v) damps oscillations and partly because the historical 5% tolerance is loose enough to accept small residual oscillations rather than chase them indefinitely. Second, the iteration is performed independently in each time step; convergence achieved in one period is not assumed to carry over to the next, since the technology stock has rolled over and the cost surface has shifted.

3.4 Discount rates

Discount rates appear in two distinct roles in CIMS, and conflating them is one of the easier mistakes to make when reading the equations in Chapter 4. The roles correspond to different *questions* about future cash flows. Being explicit about which question is being asked is what motivates CIMS to carry both a *financial* discount rate and a *behavioural* (sometimes called *revealed* or *empirical*) discount rate, and to use them in different places.

3.4.1 Financial rate

The financial discount rate is the rate at which a competently advised firm or household, applying standard cost-benefit logic, would discount the future cash flows associated with a piece of equipment. In a textbook treatment it is the weighted-average cost of capital for a firm, or the appropriate risk-adjusted social discount rate for a public project; in practice it is the rate that converts an undiscounted stream of future operating and energy costs into a present value comparable to today's capital outlay. CIMS uses this rate, denoted r, inside the capital recovery factor:

$$CRF = \frac{r}{1-(1+r)^{-N}}$$

which annualises a one-time capital cost over the equipment's lifespan N (see Chapter 4). Conceptually, the CRF is asking "what flat annual payment, discounted at r over N years, has the same present value as the lump-sum capital cost?" The right answer to that question turns on what the analyst thinks the *time-value of money* is, and so the value of r here should reflect a defensible cost-of-capital estimate. Typical values are single-digit annual percentages — a few percent above the prevailing real interest rate, with adjustments for the riskiness of the investment.

3.4.2 Behavioural rate

The behavioural discount rate, by contrast, is not a question about the time value of money at all. It is a question about *how decision-makers actually behave*. Empirical studies of consumer choice over energy-using equipment — refrigerators, light bulbs, water heaters, vehicles — find again and again that the implied trade-off between higher up-front capital cost and lower future operating cost is far steeper than any defensible financial discount rate would predict. Consumers, in the language of the literature, act as though they are discounting future energy savings at rates of 20%, 30%, sometimes higher. Firms making smaller, more reversible equipment decisions exhibit something similar, if usually less extreme.

This wedge between “what the financial calculation says they should do” and “what they actually do” is real, and it is not just irrationality. It encodes risk aversion in the face of uncertain future fuel prices, liquidity constraints, information costs, the implicit option value of not committing capital to a long-lived piece of equipment, transaction costs of switching, and decision-maker-specific tastes that the analyst cannot observe. A behavioural discount rate is the device CIMS uses to summarise this entire wedge in a single calibratable parameter. When it appears in the lifecycle-cost calculation — in particular in the foresight functions that build expected future emissions charges into today’s investment decisions (see Chapter 4) — it is the rate at which a *plausibly modelled* decision-maker, not a textbook one, is treating future costs.

3.4.3 How the two rates are used in practice

In ordinary use, CIMS applies the *behavioural* rate throughout the lifecycle-cost calculation — the foresight functions, the discounted-expectations evaluation of emissions charges, and the other places where future costs are weighed against present ones. This is the default for the reason given above: if CIMS evaluated those quantities at a textbook financial discount rate, it would predict consumers eagerly switching to high-capital-cost / low-operating-cost technology — heat pumps, LED lighting, electric vehicles — in volumes that decades of empirical observation tell us they do not, in fact, choose. The behavioural rate corrects the model toward observed behaviour without requiring the analyst to encode every individual reason for that behaviour separately.

The financial rate is still available to use in the model, but as a sensitivity and a normative reference point rather than as the default driver of technology choice. An analyst can substitute it in to test alternative views on the cost of capital, to explore how technology choice would look if decision-makers were behaving textbook-rationally, or to produce an after-the-fact cost-benefit appraisal at a defensible social discount rate.

One caveat that comes with this distinction is that model output should be read with the behavioural rates in mind. A run that holds the behavioural rate fixed across a counterfactual is implicitly assuming that the policy in question does not change the underlying barriers to adoption — and many of the policies CIMS is asked about (information campaigns, financing programmes, dealer-incentive schemes) are precisely policies that target those barriers. In such cases, the behavioural rate itself could be considered an object of policy, not a fixed parameter.

3.4.4 Behavioural rate versus intangible cost

The behavioural discount rate is not the only device in CIMS that absorbs non-financial preferences — intangible costs (described in the next section) do similar work. In an ideal world the two would carry conceptually distinct loads: the behavioural rate would capture the *time-preference* component of the wedge — impatience, liquidity constraints, the implicit option value of deferring an investment — while the intangible cost would capture the *state-preference* component — risk aversion to a new technology, perceived quality, social signalling, switching costs unrelated to time. In this idealisation the two are independent and separately identifiable from the data.

In practice they are not. Most observable patterns of technology adoption can be rationalised by any of several combinations of high discount rate and high intangible cost, and the experiments that would in principle separate the two — varying payback horizon while holding perceived risk constant, say — are difficult to run cleanly for real durable-goods decisions. CIMS handles this empirical muddiness pragmatically: it adopts behavioural discount rates from the published literature, where they have been estimated for specific decision contexts and survive across studies, and uses intangible costs as the *adjustable* calibration handle that closes the remaining gap to observed market shares (see Chapter 39). Drawing the line in this place is a choice, not a statement about which device is more "real" — it reflects the practical fact that the literature is a better source for one and historical market-share data the better source for the other.

3.5 Intangible costs as a calibration device

The lifecycle cost of a technology in CIMS is, in its short form, the sum of an annualised capital cost, an operating-and-maintenance cost, and a service cost — each of which has an obvious engineering interpretation. The full expression in Chapter 4 adds two further terms inside the capital cost: a *fixed intangible cost* (FIC) and a *declining intangible cost* (DIC_t). These are not engineering quantities, and they are not measured from invoices or fuel bills. They are calibration handles, and their role inside the model is best understood in those terms.

3.5.1 What intangible costs are doing

An intangible cost is the model's way of expressing, in dollars, everything that drives a decision-maker's preference for or against a technology *other than* the financial costs the engineering data already covers. The list of factors any one of these terms can be standing in for is long: the perceived reliability of a new technology relative to an incumbent, the quality or aesthetic of the service it provides (the warmth of incandescent light, the cargo flexibility of a pickup), the inconvenience of switching, the worry about resale value, the social signal of choosing one option over another, the perceived risk of fuel or technology lock-in. Each of these is real and each affects market share, but none of them shows up in a capital-cost or operating-cost spreadsheet. The intangible-cost terms give the model a place to put them.

Mechanically, intangible costs enter the lifecycle-cost calculation through the same channel as the engineering-derived capital cost (see Chapter 4). They are denominated in the same units, they affect the Weibit competition through the same lifecycle-cost ratio, and a high intangible cost on a technology has the same modelled effect as a high financial cost: it lowers that technology's market share in the competition. The difference is only in interpretation — financial costs are measured externally, intangible costs are inferred internally.

3.5.2 Why this is a calibration device

The defining feature of a *calibration device* is that the analyst sets its value not from independent measurement but from the requirement that the model reproduce something observable. For intangible costs, the observable target is the empirical history of market shares: in any given historical year, the actual stock-share split among competing technologies is known, and the value of the intangible-cost parameters can be adjusted until the model's competition produces the same split.

In effect, the intangible costs absorb whatever wedge remains between the engineering cost ratios — capital plus operating plus service — and the cost ratios that, run through the Weibit equation, would actually produce the observed shares. The smaller this wedge is, the closer the engineering data alone comes to explaining adoption behaviour. The larger it is, the more of the adoption story is being carried by the things engineering data cannot see. Either outcome is informative, and one of the secondary purposes of the calibration exercise is to expose where the engineering story is and is not sufficient.

CIMS treats intangible costs as a *per-competition* parameter for the same reason it treats v that way: the non-financial preferences attached to a residential furnace decision are not the same as those attached to an industrial process-heat decision. The calibration workflow that fits these parameters jointly with v against historical data is described in Chapter 39.

3.5.3 Fixed vs. declining intangible costs

CIMS distinguishes two intangible-cost terms because the underlying behaviour has two distinct timescales.

The *fixed intangible cost* (FIC) represents the time-invariant component of non-financial preference — the wedge that does not go away with experience or market exposure, because it reflects a genuine and durable feature of the technology. A user who finds gas-stove cooking irreducibly preferable to induction for reasons that have nothing to do with the price of either is not going to change their mind because more of their neighbours bought induction stoves. FIC is the calibration handle for that kind of preference.

The *declining intangible cost* (DIC_t), by contrast, represents the time-varying component — the wedge that *does* shrink as the technology becomes more familiar, more widely available, and less perceived-risky. New technologies tend to attract a risk premium that fades as adoption climbs, both because direct experience accumulates and because adopters can observe their neighbours' experience. CIMS captures this via the declining intangible-cost function (in Chapter 4), which lets DIC_t shrink as a function of new market share. This formalises the *neighbour effect* documented in the empirical literature on consumer adoption of new vehicle technologies (Mau et al. 2008) and is a key channel through which policies that boost early adoption — pilot programmes, fleet procurement, financing assistance — produce dynamic feedback in the model.

3.5.4 Limitations of intangible costs

Calibrated parameters can be made to fit almost any historical observations, and intangible costs are a flexible enough device to absorb a great deal of model error if used without discipline. Practically, the calibrated values should be *interpretable*: an intangible cost large enough to dominate the lifecycle cost should correspond to a story the analyst can tell about why the technology actually has that much non-financial baggage. A calibrated value that has no narrative is more likely to be picking up a misspecification elsewhere in the model — a wrong elasticity, a missing technology, an unrepresented constraint — than a genuine non-financial preference. Further, the values should be *stable across reasonable variations in the calibration target*: an intangible cost that swings wildly between two equally defensible historical fits is also more likely to be model error in disguise than a real preference.

Intangible costs are CIMS's primary device for expressing the gap between what engineering data says decision-makers should do and what observed adoption tells us they actually did. The calibration chapter (Chapter 39) discusses the practical workflow and the diagnostic checks that try to keep intangible costs realistic.

4 Technology Competition

Sub-model technology choice occurs according to the following order:

- Demand Assessment
- Retirement
- Assessment of capital stock availability
- Retrofit & new capital stock competition to meet demand not supplied by existing capital stock

4.1 Demand Assessment

All of the CIMS energy demand models (industry, transportation, commercial / institutional and residential) start with an exogenously determined demand for physical goods and services in each run period. These demands are mostly taken from Natural Resources Canada forecasts. The energy supply models may either use exogenously determined forecasts or they can be run using energy demand calculated from the demand model plus net exports.

4.1.1 Request-Provide demand

4.1.2 Distributed supply

4.2 Capital Stock Retirement

CIMS begins each simulation with an existing base stock of technologies. Equation 4.1 describes how the existing stock of unknown age already in use is retired. Each year a constant amount of the base stock is retired until there is no more base stock remaining (Figure 4.1).

$$BS_t = BS_0 \left(1 - \frac{T_t - T_0}{L}\right) \tag{4.1}$$

BS_t base stock at time t
BS_0 initial base stock
T_t current simulated year

T_0 initial year
L lifespan of the technology

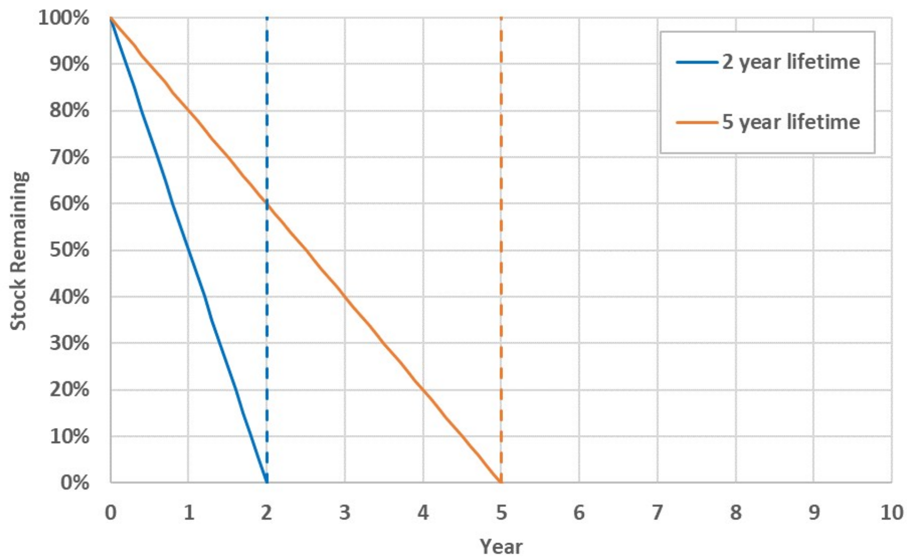

Figure 4.1: Existing Stock Retirement

Retirement of capital stock purchased during a CIMS simulation is calculated using a different methodology since the vintage of new stock is known and an explicit lifetime (and retirement schedule) can be applied to the stock purchase.

$$NS_t = \frac{NS_p}{1 + e^{\frac{11.513}{L}((T_t - T_p) - L)}} \tag{4.2}$$

NS_t new stock not yet retired at time t
NS_p new stock acquired at purchase time p
T_t current simulated year
T_p year at purchase time p
L lifespan of the technology
11.513 intercept value from ISTUM model for average technology retirement rates

This function uses an s-shaped, logistic curve to simulate the retirement of technologies. Figure 4.2 shows examples of the retirement function for technologies with different lifespans. Half of the stock is retired before the average lifespan of the technology due to early replacement and failure and half is retired afterward to simulate technologies that last longer than the average. The default intercept value of 11.513, which comes from market research on product failure rates, is used in CIMS for many technologies that are purchased and installed. For other longer-lived technologies, like building shells, lower intercepts are used that flatten the logistic curve.

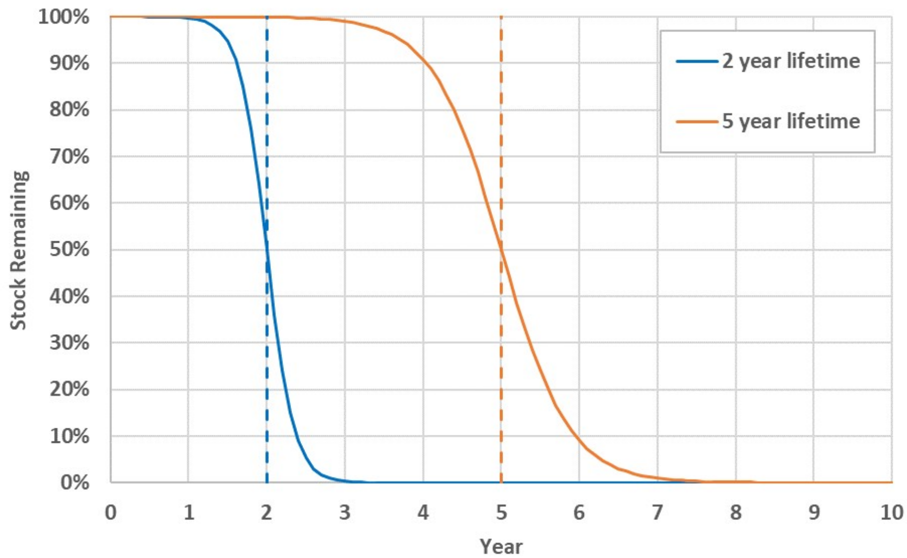

Figure 4.2: New Stock Retirement

The total stock of a technology in any given year t is the sum of the unretired base stock and the unretired new stock remaining from each previous year (Equation 4.3).

$$Stock_t = BS_t + \sum_{p=0}^{t} NS_t \tag{4.3}$$

4.3 Retrofit and New Stock Competition

While retrofitting occurs before new stock competition, the retrofitting equations are a subset of the new stock equations and are therefore presented after them. The market shares for individual technologies are determined using the equations below.

4.3.1 New Stock Competition

Equation 4.4 is a normalized inverse-power function: the market share of each technology is proportional to its lifecycle cost raised to $-v$, divided by the sum of that quantity over all competitors. It is the multinomial choice analogue of softmax, but with a power-law transform

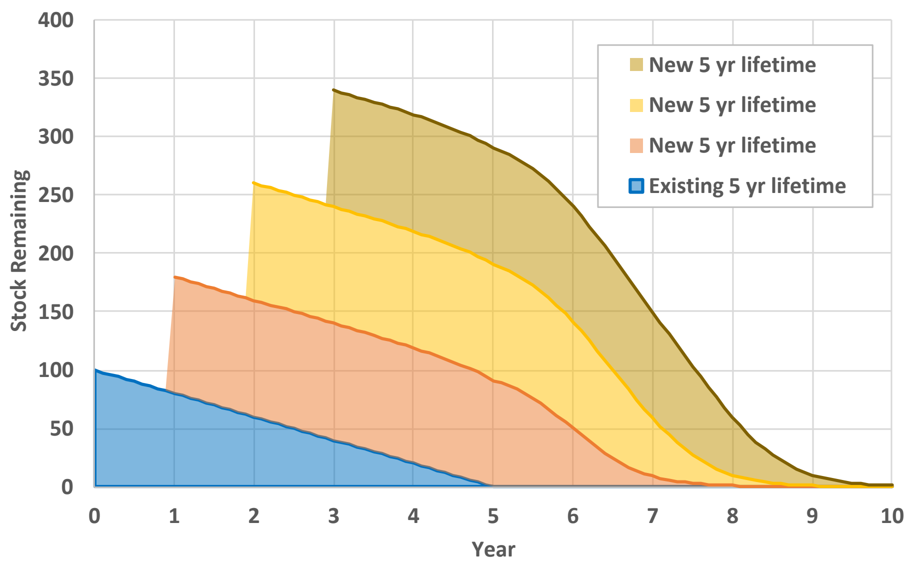

Figure 4.3: Total Stock

of cost instead of an exponential[1] — which means lifecycle-cost *ratios*, rather than differences, drive the share split. The variance parameter v controls how sharply the competition responds to those ratios: a high value, such as $v = 30$, makes the lowest-cost technology capture almost all new equipment stocks (approaching the winner-take-all rule of a linear program), while a low value, such as $v = 1$, spreads new equipment shares almost evenly across competing technologies even when their lifecycle costs differ significantly. Equivalently, in log-cost space the equation has the form of a multinomial logit on ln LCC.

$$MS_k = \frac{LCC_k^{-v}}{\sum_{j=1}^{J} LCC_j^{-v}} \tag{4.4}$$

MS_k market share of technology k for new equipment stocks
LCC_k annual lifecycle cost of technology k
v variance parameter representing cost homogeneity

[1]The CIMS choice rule belongs to the family of log-linear utility, multiplicative-error discrete choice models sometimes called the **Weibit** model. Standard multinomial logit (McFadden 1974) assumes a Gumbel-distributed additive error term in utility and yields choice probabilities of the form $e^{V_k} / \sum_j e^{V_j}$; the Weibit instead assumes a Weibull-distributed multiplicative error and yields the power-form expression used here (Castillo et al. 2008). The distinction occasionally matters when calibrating v against empirical data, because the implied stochastic interpretation of the lifecycle cost differs. For background on how the logit-style framework is used in CIMS more broadly, see Rivers and Jaccard (2005a).

$$LCC_{kt} = CC_{kt} + OM_k + SC_{kt} \tag{4.5}$$

LCC_{kt} annual lifecycle cost of technology *k* at time *t*
CC_{kt} capital cost of technology *k* at time *t*
OM_k operating costs of technology *k*
SC_{kt} service cost of technology *k* at time *t*

Note somewhere that technology capital costs should be included as CAPEX where possible to ensure apples-to-apples comparison across the competition.

$$CC_t = \max\{DCC_t, CC_{2000} \cdot DCC_{limit}\} \tag{4.6}$$

CC_t annual capital cost of a technology at time *t*
DCC_t declining capital cost at time *t*
CC_{2000} capital cost of a technology in the year 2000 (before applying the declining capital cost function)
DCC_{limit} lowest value to which a technology's capital cost can be reduced by the DCC function (as a percentage of the capital cost in year 2000)

See the section below for the declining capital cost functions.

$$CC_{2000} = \left(\frac{CC_{overnight} + FIC + DIC_t}{Output}\right) \cdot CRF \tag{4.7}$$

CC_{2000} capital cost of a technology in the year 2000 (before applying the declining capital cost function)
$CC_{overnight}$ overnight capital cost
FIC upfront fixed intangible cost, representing non-financial costs of investment
DIC_t upfront intangible cost that can decline over time
$Output$ material output of a technology
CRF capital recovery factor

$$CRF = \frac{r}{1 - (1 + r)^{-N}} \tag{4.8}$$

CRF capital recovery factor of a technology
r financial discount rate
N lifespan of the technology

$$SC_{kt} = \sum_j P_{jt} \cdot R_{kjt} \tag{4.9}$$

SC_{kt} service cost of technology *k* at time *t*

P_{jt} price of service type *j* at time *t* (can be either an energy price or the weighted-average life-cycle cost of a service, based on the market shares of technologies competing to provide that service)

R_{kjt} amount of service type *j* requested by technology *k* at time *t*

Note that market shares used to calculate the weighted-average LCC of a service are from the previous iteration of the demand-supply equilibrium process (i.e. the final market shares from the previous year if calculating the first iteration or the previous demand-supply iteration if equilibrium prices have not yet been found).

Subsidies

4.3.2 Retrofit

The retrofit option allows functioning equipment to be converted to a different type of equipment before its retirement date. For example, a boiler could be retrofitted into a cogenerator before it's lifetime-based retirement. After existing stock is retired, CIMS treats the initial purchasing price of the incumbent technology as a sunk cost and uses the operating and maintenance (O&M), energy, service and emissions costs as the only costs of the incumbent technology. The costs of the retrofit options are then calculated as the sum of O&M, energy, service and emissions costs plus the annualized retrofit capital cost less any tax or other credit. CIMS then determines, using a similar competition calculation to those already described, how many incumbent technologies will be retrofitted.

The analyst can place some bounds on this retrofit option. A maximum (R-MAX) or minimum (R-MIN) market share can be established such that only a certain amount of the old stock could (maximum) or must (minimum) be retrofitted in any year. In an unconstrained market, maximum should be set to 1 and minimum to 0.

4.4 Modifications to the technology choice equations

Several functions have been added to the competition process to reflect both real world conditions and policy concerns. These currently include functions to:

- Reduce capital costs as total production increases to reflect learning by doing and efficiencies of scale;
- Reduce intangible costs as new market share climbs to reflect reduced perceived risk (i.e. the neighbour effect); and
- Eliminate obsolete but still functioning technologies.

4.4.1 Declining Capital Cost

The declining capital cost function allows the capital costs of technologies to decline as a result of learning by doing and improving economies of scale. In CIMS, capital costs decline as a result of the cumulative production of a technology across Canada and an exogenous rate of decline which represents production in other countries.

Need to align this description and function below with Thomas' update of DCC function

$$DCC_t = GCC_t \cdot \left[\left(\sum_{1}^{p} \frac{CumulNS_{2000,p} + \sum_{j=2005}^{t-5} NS_{jp}}{CumulNS_{2000,p}} \right)^{\log_2(PR)} \right] \tag{4.10}$$

DCC_t declining capital cost of a technology at time t

GCC_t capital cost of a technology adjusted for cumulative stock in other countries

NS_{jp} new stock of a technology added from 2005 to the previous time period in each province p

$CumulNS_{2000,p}$ cumulative new stock of a technology for all years up to and including the year 2000 in each province p (exogenous)

PR progress ratio

Increased stock in other countries reduces the capital costs of a technology in CIMS. Therefore, even if there is no change in stock of a technology in Canada, its capital costs may still decline. The rate at which the capital costs decline as a result of global stock change is set exogenously.[2]

$$GCC_t = GCC_{t-5} \cdot (1 - AEEI)^5 \tag{4.11}$$

GCC_t capital cost of a technology after it has been adjusted for cumulative stock in other countries

$AEEI$ exogenous rate of decline of a technology's capital costs

In order to prevent capital costs from declining indefinitely, an analyst may specify the percentage of the initial capital costs below which capital costs may not decline.

Research has shown that investment in one technology (e.g., hybrid cars) can lower costs of another technology (e.g., battery cars). This can be achieved by setting technologies to the same DCC category so that the sum of all installed stock for technologies within that category is used when determining capital costs.

[2] This is equivalent to the autonomous energy efficiency index (AEEI).

4.4.2 Declining Intangible Cost Function

The declining intangible cost function allows non-financial costs to fall as a result of reduced perceived risk as market exposure increases. The following general formula applies.

(Equation 20 — placeholder for the declining intangible cost function) (4.12)

I_t intangible cost in time t

I_0 initial intangible cost (column N of the economic file for capital, column R for operating costs)

A constant in column O of the economic file for capital, column S for operating costs

K constant in column P of the economic file for capital, column T for operating costs

NMS new market share

NMS_{t-5} new market share from the previous run, expressed between 0 and 1 (i.e. between 0% and 100%, not 0 and 100). NMS is the technology's share among all other technologies that provide the same service.

4.4.3 Forward looking behaviour

The foresight functions simulate business and consumer behaviour for considering future costs. The functions calculate the expected emissions cost of technologies, considering different lifespans and discount rates.

CIMS has three options for generating expectations: 1) myopic, where people do not have foresight into their future emissions costs; 2) foresight based on the average emissions costs of a technology; and 3) foresight based on the discounted value of emissions costs of a technology.

Myopic

The emissions charge used to calculate the lifecycle cost of technology is the current tax rate.

$$E(\text{EmissionsCharge})_{kt} = \text{EmissionsCharge}_t \tag{4.13}$$

Average Expectations

The second option generates expectations of future emissions costs based on the average emissions charge over the lifespan of a technology.

(Equation 22 — placeholder for the average-expectations function) (4.14)

$\textbf{EmissionsCharge}_n$ emissions charge in year n, set in CIMS's Simulation Creation Centre

$base$ year in which technology k is purchased

N lifespan of technology k

Discounted Expectations

The third option generates expectations of future emissions costs based on the present value of emissions charges. After the present value of emissions charges has been calculated, it is annualized using the capital recovery factor.

(Equation 23 — placeholder for the discounted-expectations function) (4.15)

EmissionsCharge$_n$ emissions charge in year n, set in CIMS's Simulation Creation Centre
$base$ year in which technology k is purchased
N lifespan of technology k
r_k revealed discount rate of technology k, set in the Techno Economic file

5 Energy Supply and Demand

5.1 Energy Pricing

5.1.1 Primary Energy

5.1.1.1 Fixed Price

5.1.1.2 Cost Curve – Annual

5.1.1.3 Cost Curve – Cumulative

5.1.2 Secondary Energy

5.1.3 Carbon pricing

TODO: Add section on biomass supply modelling

In CIMS's integrated energy supply and demand system, the demands of final demand sectors directly drive the activities of electricity production, refining, coal mining and natural gas and crude oil extraction. Inter-regional transfers as well as net exports are added, as demand for Canada's energy supply sector includes US demand. The importance of domestic demand varies by energy commodity. Most electricity is produced for Canadian consumers, more than half of natural gas produced is shipped to the US, and there is a heavy trade in crude oil and refined petroleum products. Half our domestic crude oil production is shipped to the US, while we import from both OPEC and non-OPEC suppliers.

Need to revise this description to reflect the current setup, but should preserve a more complete version as a comment for later upgrades. Adjust the descriptions below to detail the fully endogenous options and note that many can be switched to exogenous function.

Each of the energy markets in CIMS has the option of fixed or dynamic pricing, and they are described below:

- Regional electricity markets are included with endogenous internal pricing based on the cost of production. The amount of electricity produced is the sum of endogenous domestic demand and net extra-regional exports, which may be adjusted using an export own-price elasticity.

- One national natural gas market is included, with dynamic pricing using a supply curve based on a combination of Canadian and US market feedbacks. The amount of gas produced is the sum of endogenous domestic demand and net exports. Net exports may be adjusted to reflect changes in the domestic cost of production using an export price elasticity.

- One national crude oil market, based on an exogenously specified world price of crude oil. The amount of crude oil produced is the sum of endogenous domestic demand and net exports. Net exports may be adjusted to reflect changes in the domestic cost of production using an export elasticity. The world crude oil price is assumed to be unresponsive to Canadian demand.

- Petroleum refining regions divide their production amongst demand regions and net exports. All can use endogenous internal pricing based on the cost of production. The amount of refined petroleum products produced is the sum of domestic regional demand and net extra-regional exports.

- Coal producing regions divide their thermal coal production amongst endogenous domestic regional demand and exogenous exports (i.e. BC exports 100% of its production while Alberta exports about 30% of its production). Due to the differences between the ways each non-producing region and sector gets its coal, these regions are assumed to buy coal at a fixed market price or according to a local supply curve – for example, some electricity plants in Alberta and Saskatchewan are sited at coalfields and are the sole customers. British Columbia, Alberta, Saskatchewan and the Atlantic provinces all buy their coal at a fixed market price.

- **-> Revise coal description to separate thermal coal (exogenous) and metallurgical (endogenous).**

CIMS divides energy flow in the economy into demand for energy end-use services, intermediate processing, and primary supply (Figure 5.1). It begins with an initial demand for four general end-use energy commodities: electricity, natural gas, coal, and refined petroleum products. The electricity needs of the domestic energy demand models (transportation, residential, commercial and industry) plus net electricity transfers (inter-regional transfers and exports) are used to drive electricity generation. The refined petroleum products sector is driven by the net demand from the energy demand models, electricity, inter-regional transfers, and net exports. The primary energy supply models, crude extraction, coal mining and natural gas extraction and transmission are driven by the net demand for their products from the final demand sectors, electricity and refined petroleum.

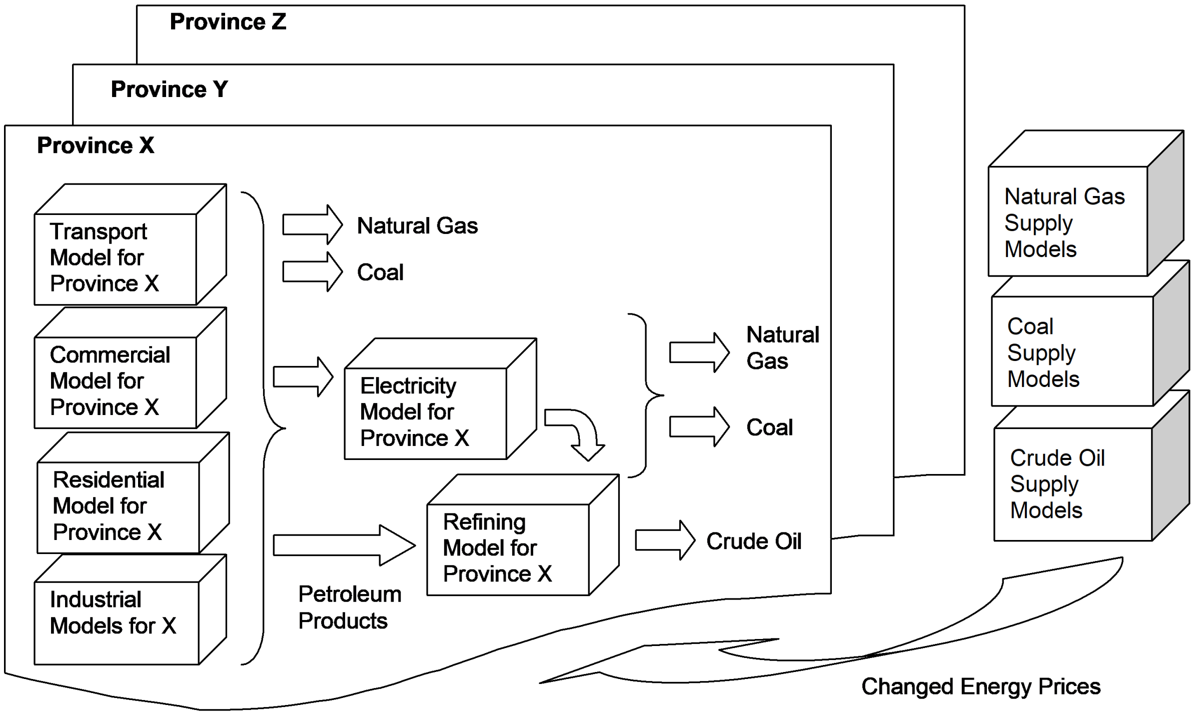

Figure 5.1: final_intermediate_primary_flow

The electricity demands of petroleum refining, the refined petroleum product demands of the primary energy supply models, and the interdependencies among the primary energy supply models create feedback loops within the system. Because CIMS operates as a simulation model rather than a computable general equilibrium model, these feedback loops cannot be resolved within a single iteration of the demand–supply algorithm. To address this limitation, CIMS is run iteratively until lifecycle costs throughout the model converge below a specified threshold criterion (typically less than a 5% difference between iterations). Additional iterations may also be performed to ensure that the effects of the feedback loops have stabilized.

Canada trades a significant amount of energy with the United States and, to a lesser extent, with both OPEC and non-OPEC suppliers. Given the possible effect of Canadian energy policy on the price of producing these commodities, and hence their marketability to the US, electricity, crude oil and natural gas export feedbacks are included in CIMS using price elasticities calculated for the Canadian Federal Department of Finance (Table 5.1).

change this table to GTAP? talk to Thomas

Table 5.1: Price Elasticities for Energy Exports, Wirjanto (1999)

Energy Commodity	Long-Run Own-price Elasticity for Export Demand
Electricity	-0.54
Natural gas	-0.67
Coal	-0.51
Gasoline and fuel oil	-0.92
Other petroleum products	-0.82

These elasticities follow the Armington specification, whereby the demand for an imported good rises as its price falls relative to the price of its domestic substitute. At all times, consumers buy at least some of the domestic good and some of the imported good, even when the price difference is large (Iorwerth et al. 2000, Wirjanto 1999). See Bataille (2005) for further elaboration. The Armington specification assumes that goods produced by different countries are not perfect substitutes. The energy trade elasticities are represented in Equation 5.1, which applies the elasticity of the ratio of import volumes to domestic production in relation to a change in the relative price of imports to domestic prices.

(Equation 25 — placeholder for the energy trade Armington elasticity) (5.1)

$FLCC_t$ current financial lifecycle cost for producing the energy commodity
$FLCC_{t-1}$ previous financial lifecycle cost for producing the energy commodity
COP cost of production covered by the energy-producing sub-model, as a percentage of total cost
σ price demand elasticity for the energy commodity
IT initial trade volume of the energy commodity
Adj_{prod} additive adjustment to production of the energy commodity

Western Canadian exports of crude oil are separated from eastern Canadian imports, as there are currently no large cross-country shipments. While electricity trade is currently a very small portion of our energy trade, electricity import substitution elasticities are included to cover the possibility of a policy driven increase or decrease in electricity exports to the US.

6 Goods and Services Demand Feedbacks

The economy is commonly divided into: households, who provide labour and financial capital and who consume goods and services; firms, who consume capital and labour and provide goods and services; government, which consumes goods and services, provides public goods and regulates the market; and finally foreign producers and consumers of goods, services and capital (Dornbusch et al. 1989). It is common to assume that Canada is a price taker for capital (i.e., the interest rate is set at the world price of capital).

> Canada is assumed to be a small open economy... the world interest rate is assumed to be the US prime rate, and foreign savings are assumed to be available elastically. This means that the foreign sector automatically clears any surplus or deficit in the domestic savings market each period, so the domestic interest rate is effectively exogenous. (McKitrick 1998) p.552

The following suite of options face the consumer or producer when the price of a good or input rises, modelled after the consumer's utility map in CASGEM (Iorwerth et al. 2000):

- Substitute an import
- Buy something else
- Substitute current consumption for savings
- Work less and enjoy more leisure, which is the same as reducing output

These options were used as the primary guide for adding demand feedbacks in CIMS.

6.1 Manufactured products

Most Canadian industrial products are highly traded; their sensitivity to foreign competition is of prime importance. To reflect the sensitivity to foreign competition, CIMS employs the same Armington specification for price elasticities as used for energy trade, whereby the demand for a foreign good rises as its price falls relative to the price of a domestically produced substitute. Using the Armington specification, consumers always buy at least some of the domestic good and some of the imported good. The price elasticities are applied with the general formula in Equation 6.1.

(Equation 26 — placeholder for the manufactured-products Armington elasticity) (6.1)

$FLCC_t$ current financial lifecycle cost for the product
$FLCC_{t-1}$ previous financial lifecycle cost for the product
COP cost of production covered by the sub-model that produces the product, as a percentage of total cost
σ price demand elasticity for the product
$Multiplier$ value by which demand for the product is multiplied

To accommodate the differences between products in industrial sectors, especially Industrial Minerals, Chemical Products, Metal Smelting, and Light Industrial, demand feedbacks are applied at the level where production of individual products is determined. Table 6.1 provides the elasticities used for sectors and their products. See Appendix E of Chris Bataille's PhD dissertation for a full breakdown of their sources and how they were mapped to CIMS' sub-models and their sub-products.

Table 6.1: : Price elasticities used from Wirjanto (1999)

Sector	Product	Elasticity
Metal Smelting	Lead, Zinc, Aluminium, Nickel	-0.59
	Copper	-0.57
Chemical products	Chlor-alkali, Sodium Chlorate, Hydrogen Peroxide, Ammonia Methanol	-0.44
	Polymers	-0.83
Industrial Minerals	Cement	-1.75
	Lime, Glass, Bricks	-0.53
Pulp and Paper	Pulp export	-1.72
	Coated	-0.79
	Uncoated, linerboard, tissue & newsprint	-1.54
Iron and Steel	Slabs, blooms, billets & molten steel	-0.60
Mining	Open pit, underground	-1.38
	Potash	-0.58
Light manufacturing	Food, tobacco, beverage	-0.32
	Leather, textile, clothing; Wood products	-0.86
	Rubber and plastics product	-1.30
	Electrical and electronic equipment	-0.81
	Transportation equipment	-1.06
	Furniture, printing, machinery	-0.74

6.2 Commercial and residential

Check this section because I don't think we do this! Review with Thomas how much of Chris Bataille's work is actually included. Also need Thomas to document his macroeconomic work.

There is no import substitution for non-traded goods and services such as residential and commercial floorspace. CIMS uses a log-linear econometric relationship to estimate the relationship between manufacturing activity – used dynamically as valued added but proxied using the value of shipments – and residential / commercial activity – measured by the value of building permits. The resulting relationship is used to allow commercial and residential activity to respond to changes in manufacturing activity.

(Equation 27 — placeholder for the commercial/residential demand feedback) (6.2)

$BP_{c,r}$ natural log of activity (value of building permits) in the commercial or residential sectors in a given year

IVA natural log of industrial value-added in a given year

e error term

m coefficient of the relationship (elasticity)

b intercept

This equation was estimated, using SHAZAM v.9, by running ordinary least squares regressions between manufacturers' shipments and residential building permits between 1961 and 1991 and manufacturers' shipments and commercial, institutional, and governmental building permits between 1961 and 1991. All values were deflated using a standard consumer price index. We tested various permutations of the regressions, including lagging and the use of population and year as an independent variable. Table 9 provides the parameter estimates.

Table 7. Parameter Estimates Used to Provide Demand Feedbacks for the Residential and Commercial/Institutional Sectors

The process for calculating manufacturing valued-added, the driver for residential and commercial activity in CIMS, is complex. The sub-model's outputs are designed to capture changes in energy, new capital and labour expenditures engendered by policy. These costs are a sub-component of the total cost of production, which also includes materials and supplies, as well as capital investment, labour and repairs not related to energy using equipment.

Value-added is defined as the total value of shipments and other revenue, minus the cost of intermediate inputs (energy, materials, and supplies) (Lipsey et al. 1998, p.175). Returns to capital include both amortization and profits. Using values for wages, energy expenditures, costs of material, total value of shipments and capital expenditures from Statistics Canada, we found that while the ISTUM sub-models do not capture expenditures for materials or capital unrelated to energy consumption, they capture most capital expenditures in the sectors for which sub-models exist. Supplemental to this, the demand feedback function of CIMS, when faced with a reduction in demand, removes first new growth, then the necessity for replacing retired technology stock, and finally useful stock that is no longer required. Given that most sectors in CIMS grow in most time periods, a simplifying assumption was made that all the productive investment being removed was new or replacement investment, not useful stock. We then compared CIMS' new investment in 2000 with figures from Statistics Canada, and derived a set of multipliers to scale up CIMS' investment figures to reflect the ratio of coverage (the multipliers affect the demand for all stock, not just the portion of the cost of production covered by ISTUM). Finally, we added normal profits as a percentage of revenue, having ascertained that normal profits are included in value-added and that these are not normally included in CIMS. These were calculated as a percentage of total costs from Statistics Canada's "Financial and taxation statistics for enterprises, 2000, 61-219".

6.3 Freight transportation

Freight transportation is important for energy and emissions policy because of the large amount of energy consumed and the amount of GHGs and criteria air contaminants produced. Activity in this sector is directly related to the amount of activity in the rest of the economy. To provide feedback relating freight and economic activity, a similar relationship to that established between manufacturing value-added and residential and commercial activity was incorporated into CIMS. It proved difficult to find an equivalent statistic for freight activity to manufacturing activity as used for the commercial and residential sectors, so instead of an estimated relationship a rule-of-thumb value was temporarily used instead (0.95 – a 1% fall in manufacturing value-added causes a 0.95% fall in freight activity). This relationship allows changes in manufacturing value-added to adjust freight transportation activity. This relationship will be estimated once statistics are available.

6.4 Personal transportation

Demand feedbacks for personal transportation operate in a similar fashion to the demand feedbacks for manufacturing and industry, except that in the place of an Armington price elasticity an elasticity for personal kilometers traveled (pkt) in response to the cost of travel is used instead (price elasticity =-0.02) (Michaelis and Davidson 1996). Transportation is also a bit different from the other sub-models in that some demand adjustments occur within the sub-models, specifically the choice between urban modes of transport (single occupancy vehicles, high occupancy vehicles, transit and walking and cycling). The upshot is that much of the structural change possible in transportation is directly included in the sub-sector models.

6.5 Limits to demand adjustments

Experimentation with early versions of CIMS showed that the demand adjustments had to be regarded critically. Elasticities are by nature log-linear measurements, while demand responses are very often non-linear. It may be hypothesized that there will always be a demand for some commodities, no matter the price. A function for limiting their effects was installed; 50% was chosen as the maximum demand reduction as a rule-of-thumb for the maximum extent to which CIMS' substitution elasticities were still applicable. This demand limiting adjustment is somewhat simplistic, and in the future some form of function that applies limits to demand adjustments in a progressive way will be installed.

Part III

Sector Descriptions

7 Sector Overview

TODO: add sections on renewable fuel blending, cogeneration, waste heat recovery.

In CIMS, the product and energy service demands in a sub-sector are linked in a flow model that describes the sequence of activities required to generate that product. A CIMS flow model is geared towards representing technology evolution and energy consumption rather than economic criteria (as in an econometric model where units are typically in monetary terms) or actual mechanical processes (as in the blueprints or process flow diagrams used by engineers). Because the emphasis is on energy consumption and not material flow, the nodes in the flow model represent process stages in which energy consumption can be distinctly estimated. The flow model describes hierarchical nodes, linked by engineering ratios. Technology competitions take place at the lowest level nodes in the hierarchy.

Each sector sub-model is represented as a hierarchical tree (implemented in the model code as a NetworkX directed tree). Each node carries a competition rule that determines how demand passed down to it is allocated among its children:

- **Fixed-ratio** nodes split demand among their children in fixed, exogenously specified proportions (engineering ratios). These represent the unavoidable sequence of process steps required to produce the sector's output.
- **Tech-compete** nodes host a competition among alternative technologies that can provide the same service. Market shares are assigned by the standard CIMS technology-choice rule described in Chapter 4, based on the technologies' lifecycle costs and a heterogeneity (variance) parameter.

At the top of each sector tree, the nodes are organised into two sub-trees — a **Product** sub-tree and a **Service** sub-tree:

- The **Product** sub-tree is the physical production chain: the sequence of process stages that converts raw inputs into the sector's saleable output (tonnes of steel, square metres of floor space, vehicle-kilometres, and so on). It is driven by the sector's exogenous output or activity forecast, and its shape is specific to each sector's process flow.
- The **Service** sub-tree holds the auxiliary energy and utility services that production technologies request rather than producing themselves — for example space heating and cooling, lighting, machine drive, steam and cogeneration, fuel blending, and carbon capture where applicable. Factoring these shared loads into their own sub-tree avoids re-specifying the same equipment inside every process stage and lets efficiency and fuel-switching measures compete once and serve the whole sector.

For the industrial sectors, the auxiliary services follow the end-use structure of the U.S. Energy Information Administration's Manufacturing Energy Consumption Survey (MECS) — in particular, machine drive is disaggregated into compressed air, fans, materials handling, materials processing and pumps. Each service is represented at a general level, as a competition between existing and more efficient equipment with fuel-switching options where relevant, rather than as a detailed equipment catalogue. The specific set of services varies by sector; the buildings and transportation sectors, for instance, carry their own service definitions.

8 Coal Mining

TODO: Add a description of the Coal Mining sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

9 Natural Gas

TODO: Add a description of the Natural Gas sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

10 Petroleum Crude

The petroleum crude industry in Canada includes:

- A conventional crude oil production system, which comprises all facilities and associated infrastructure used to extract, treat, and transport light, medium, and heavy crude oils to market.
- An unconventional crude oil system, which comprises all facilities used to extract crude bitumen from oil sands and subsequently upgrade them to synthetic crude oil. It is disaggregated into in-situ bitumen recovery and oil sands mining/upgrading.

10.1 Conventional crude oil

The production, treatment and transmission segment of the conventional crude oil system may comprise the following infrastructure:

- Oil wells which produce at least one barrel of crude petroleum oil to each 100,000 cubic feet of natural gas.
- Oil transmission pipelines or tanker vehicles to transport the well effluent to a battery or a cleaning plant.
- Batteries where effluent from wells is separated into its constituent phases for metering and appropriate disposition.
- Oil storage tankers.
- Oil transmission pipelines or tanker vehicles to move crude oil from producers to market.

10.2 Oil sands mining/upgrading

Oil sands mining/upgrading usually comprises the following steps:

- Surface mining: Since the 1920's, open pit mining has been central to oil sands development. Mine equipment from the early years was scaled up significantly when large commercial operations started to come on line. In the past, draglines and bucket-wheel reclaimers were used to mine oil sands. This practice has largely been replaced by shovel-and-truck mining, which gives greater flexibility. Run-of-mine oil sands, dug directly from the open pit faces, is hauled to in-pit crushing stations, and is then slurried and pumped to the bitumen extraction plants.
- Bitumen extraction: The oil in oil sand is called bitumen, a complex hydrocarbon made up of a long chain of molecules. Bitumen is usually separated from oil sands using a warm-water extraction process. The mixing that takes place during slurry transport from the mine to the plant is sufficient to begin the separation process, which recovers over 90% of the bitumen resource. The resulting bitumen froth then forms the feedstock for creating synthetic crude oil.
- Raw bitumen upgrading: In order for bitumen to be processed in refineries, the complex hydrocarbon chain in bitumen must be broken up and reorganized, which means removing some carbon while adding additional hydrogen to make more valuable hydrocarbon products by upgrading. This is done using four main processes: coking removes carbon and breaks large bitumen molecules into smaller parts, distillation sorts mixtures of hydrocarbon molecules into their components, catalytic conversions help transform hydrocarbons into more valuable forms and hydrotreating is used to help remove sulphur and nitrogen and add hydrogen to molecules. Further treatment and blending produces light low-sulphur (sweet) synthetic crude that is shipped by pipeline to refineries.

10.3 In-situ Bitumen

About 80% of the oil sands in Alberta are buried too deep below the surface for open pit mining. This oil must be recovered by in situ techniques ("in situ" is Latin for "in place"). Using drilling technology, steam is injected into the deposit to heat the oil sand lowering the viscosity of the bitumen. The hot bitumen migrates towards producing wells, bringing it to the surface, while the sand is left in place.

The most promising thermal recovery technology for Canadian bitumen resources is the steam-assisted gravity drainage (SAGD) process. In this process, two horizontal wells aligned in a vertical plane are drilled near the bottom of the reservoir. The top horizontal well is used to inject steam, while the bottom well is used to collect the produced liquids (oil, condensed steam, formation water). The steam injected from the top well rises into the formation, forming a large steam chamber above the well. The rising steam condenses on the boundary of the chamber, heating the oil and allowing it to drain to the production well. The process leads to high oil rates and a high recovery of the original oil in place at economic oil-to-steam ratios.

The SAGD process is energy intensive. For some reservoirs, the vapour extraction (VAPEX)rocess may have more potential. It is an analogue to SAGD, involving the injection of vaporized solvents (e.g. ethane, propane, butane) or solvent/gas mixtures rather than steam. This process results in the formation of a vapor chamber rather than a steam chamber, but with a similar reduction in oil viscosity at the chamber boundary that allows the oil to drain to the production well. Although VAPEX is less energy intensive than SAGD, production rates are also lower.

In addition, hybrid steam/solvent processes are currently under development for reservoirs in which steam or solvent processes alone are not suitable.

Production from in situ already rivals open pit mining and in the future may well replace mining as the main source of bitumen production from the oil sands. Challenges facing in situ process are efficient recoveries, management of water used to make steam, and co-generation of all (otherwise waste) heat sources to minimize energy costs.

Figure: Oil sands

11 Petroleum Refining

This industrial sector includes establishments primarily engaged in manufacturing petroleum products. Activities such as the manufacture of lubricating oil and grease or asphalt and coal/coke products are included in CIMS where they are carried out in these plants. Establishments that are primarily engaged in these activities are not included; however, because of their comparatively small size.

The basic process involves refining crude oil into a number of products including: ethane, propane, butane, motor gasoline (typically 40% of total product), naphthas, jet fuel, petrochemical feedstocks, distillate fuel oil, diesel fuel, residual fuel oils, lubricants, coke and asphalt. All crude includes these products in varying concentrations. However, it is possible to increase output of more desirable products per unit crude input by using cracking and reforming processes.

The type and quality of crude and the processing requirements to generate the end products determine the refinery's complexity and have significant impact on energy consumption in the plant. In spite of this, however, the basic processing procedures are common to all refineries. The major process steps include: atmospheric distillation, vacuum distillation, cracking, desulphurization, alkylation, isomerization, reforming, sulphur recovery and hydrogen production. The last two processes should not be considered part of the actual refining process; they act as utility functions to clean out sulphurous compounds or provide hydrogen for specific in-plant operations.

Figure: Energy Flow Model for the Chemical Products Industry

12 Electricity

The electricity model in CIMS is structured to account for utility-generated electricity. Electricity co-generated by industry is accounted for separately in the industrial sub-models. Please note that all electricity generated from renewables is accounted by the electricity model no matter whether it is generated by a utility or an industrial establishment.

Final electricity demanded in the entire Canadian economy (excluding co-generated electricity by the industry) is represented by the primary node (Electricity). The amounts of electricity allocated to intermediate and competition nodes flows directly from this primary node, which is the principle driver of the simulation runs. Intermediate nodes in the Electricity sub-model reflect the specific characteristics of electricity demand. Electricity demand varies significantly reflecting the fluctuating demand requirements depending on time-of-day (i.e. daily demand typically peaks between 4-6pm) and by time-of-year (i.e. annual peak demand during the cold winter months). Utilities track these varying demand conditions using load curves in an effort to supply electricity on a continual basis.

Electricity generation technologies have traditionally been classified based on their relative variable costs and the resulting frequency with which they are called upon to meet electricity demand or ''load''; e.g.,''baseload,'' ''load-following,'' and ''peaking'' resources. This classification is no longer meaningful in power systems with substantial penetration of wind and solar energy, since dispatch of each technology is also driven by the irregular variability of these renewable resources. Moreover, most available low-carbon technologies are capital intensive and have very low variable costs. In this context, the distinguishing attributes of electricity technologies relate more to their resource availability and ability to adapt production output in order to meet instantaneous demand.

1. ''Fuel-saving'' variable renewable energy (VRE) resources. These include wind power, solar photovoltaics (PV), concentrating solar power, and run-of-river hydropower. They harness renewable energy inputs (wind, solar insolation, water) that vary on timescales ranging from seconds to hours to seasons, have zero (or near-zero) variable costs, and have no fuel costs. At lower penetration levels, they may displace the need for firm capacity, but, at higher shares, capacity needs are driven by periods with low VRE availability. At high energy shares, these technologies therefore contribute value primarily by displacing other higher variable cost generating technologies whenever available and reducing the total fuel consumption and variable costs of the system.

2. ''Fast-burst'' balancing resources. These include short-duration energy storage (e.g., Li-ion batteries), flexible demand (or schedulable loads), and demand response (or price-responsive demand curtailment). They are either energy constrained (storage, demand flexibility) or have very high variable cost (demand curtailment). These technologies are therefore poorly suited to operating continuously over long periods of time and are better used during high-value periods when relatively fast bursts of power or quick demand adjustments are needed to balance supply and demand.

3. ''Firm'' low-carbon resources. These are technologies that can be counted on to meet demand when needed in all seasons and over long durations (e.g., weeks or longer) and include nuclear power plants capable of flexible operations, 12–15 hydro plants with high-capacity reservoirs, coal and natural gas plants with CCS and capable of flexible operations, geothermal power, and biomass- and biogas-fueled power plants.

Aggregating and rearranging the daily load curves in descending order from highest demand (peak) to lowest demand yields a load-duration curve (LDC). An LDC relates the capacity required with the capacity utilized, in other words the number of operating hours required during the year for each proportion of peak capacity demand. The LDC is divided into time segments (i.e. hourly or daily) with the extreme left side representing the province's peak demand for the year (i.e. usually a winter evening) while the extreme right side represents the baseline power demand (i.e. the minimum amount of power required). Interpretation of the LDC allows for the determination of capacity i.e. represented by the height of each slice, and of the utilization rate i.e. the width of each slice. The product of these two provides'a measure (estimated) of electrical energy (e.g. kilowatt-hours).

As illustrated by the energy flow model of the electricity industry, total electricity generation in CIMS is split between the base, shoulder and peak nodes using ratios based on historical load curves. These ratios can change over time but are always user-specified.

Baseload is a utility's minimum load over a given period. A baseload generating facility provides the basic amount of electric power needed year-round except for maintenance and scheduled or unscheduled outages; normally a large, efficient power plant having a low cost per kilowatt-hour generated. The base node includes conventional technologies such as gas turbines, combined cycle gas turbines, simple cycle coal turbines, pulverized fluidized bed coal, integrated gasification combined cycle coal, small and large hydro, nuclear; and renewable technologies such as biomass steam plant, fuel cell generator, parabolic trough solar generation, geothermal, solar thermal and wind generation.

The shoulder and peak nodes only contain conventional technologies. Moreover, as a peak node plant needs to be phased in and out of the grid relatively quickly, plants with long start time such as nuclear plants could not be used as peak node facilities

While the split among base, shoulder and peak nodes are always user-specified (decision rule: fixed_matrix_ratio), the split between conventional and renewable generation in the base node can be either user-specified over time or based on the aggregate relative annual levelized costs of the technologies on each sub-node.

The current CIMS model deals with electricity imports and exports through the macroeconomic module. The own price elasticity for electric power export demand is -1.07. And the energy trade elasticities are applied using the following equation:

(Equation 31)

Where:

FLCCt = Current financial lifecycle cost for a product

FLCCt-1 = Previous financial lifecycle cost for a product

β = % of market price covered by sub-model

σ = Own price demand elasticity for the energy commodity

γ = Corrected demand multiplier for the given market

Figure: Energy Flow Model of Electricity Generation

Rates

13 Ethanol

TODO: Add a description of the Ethanol sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

14 Biodiesel

TODO: Add a description of the Biodiesel sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

15 Hydrogen

TODO: Add a description of the Hydrogen sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

16 Mining

This chapter describes how the mining sector is represented in CIMS. As with the other industrial sub-models, the emphasis is on the structure of the energy-service flow model — the hierarchy of nodes, the technologies that compete at each node, and the engineering ratios that link them — rather than on the geology or economics of individual mines. The model tree is region-agnostic: a single master structure is instantiated for every province and territory, and regional differences are expressed through which branches carry production and through the data attached to each node, not through separate per-region structures.

16.1 Scope

CIMS represents both **metal mining** and **non-metal mining**. Metal mining is split into open-pit and underground operations, and within open-pit into iron and non-iron ores. Non-metal mining is dominated, in energy terms, by **potash**, which is modelled in full; other non-metal commodities are lower-energy but are within the scope of the sector and are noted below where they are not yet built out. The driving variable for the sub-model is the physical throughput of ore, expressed in tonnes; a forecast of ore throughput drives the demand for every downstream process and energy service, and is set exogenously in the same way as the output of the other industrial branches.

Several activities that are colloquially called "mining" are deliberately handled elsewhere in CIMS to avoid double-counting energy and emissions:

- **Coal mining** is covered in the Coal Mining chapter and the energy-supply component.
- **Oil and gas extraction** and **oil sands mining** are covered under Petroleum Crude.
- **Cement, lime, glass, brick and other non-metallic *mineral products*** are covered under Industrial Minerals — these are downstream manufacturing branches, distinct from the extraction of the non-metal ores treated here.

The smelting and refining of mined concentrates is likewise excluded and handled in the Metal Smelting and Refining chapter; the mining sub-model stops at the mine gate, with a saleable concentrate or product as its final output.

16.2 Real-world context

Canada is one of the world's leading mining nations, and the industry is highly regionalised. Metal mines produce base and precious metals — copper, nickel, zinc, gold, iron ore, and others — from both open-pit and underground operations. Saskatchewan is the world's largest potash-producing region, accounting on the order of 30% of global potash production (Natural Resources Canada 2025), and also supplies roughly a quarter of the world's uranium from the high-grade deposits of the Athabasca Basin (World Nuclear Association 2025). The Northwest Territories host Canada's diamond industry, and Nunavut has growing gold and iron-ore production. Mining is capital-intensive and strongly export-oriented, so activity levels are driven by world commodity prices and ore grades rather than by domestic demand — which is why CIMS drives the sector with an exogenous throughput forecast rather than an endogenous demand response.

Mining is energy-intensive, and the energy is concentrated in a few activities. **Comminution** — the crushing, grinding and milling that reduce run-of-mine rock to particles fine enough for mineral separation — is typically the single largest energy consumer at a metal mine; published estimates put it at roughly a quarter of final energy at an average mine site and a majority of mine *electricity* use (Coalition for Eco-Efficient Comminution 2021). **Diesel** dominates mobile equipment, particularly haulage, and is the largest single fuel at many sites; many remote northern mines are off-grid and rely heavily on diesel generation for all loads. **Ventilation** is a large and distinctively underground electricity load, since deep mines must move enormous volumes of air for air quality and temperature control. The CIMS structure is built around exactly these distinctions: it separates open-pit from underground mining, breaks the ore-processing chain into its comminution stages, and represents ventilation, fans, pumping and the other utility loads through the shared machine-drive service (see the Industrial Machine Drive Systems chapter).

16.3 Model structure

Following the standard CIMS pattern, the sector divides into two sub-trees: a **Product** sub-tree, which is the physical production chain that turns ore into a saleable product, and a **Service** sub-tree, which holds the auxiliary energy and utility services that the production technologies draw on. Figure 16.1 shows the overall structure.

As described in the Sector Overview, each node carries a competition rule. **Fixed-ratio** nodes impose the unavoidable engineering sequence of process steps, while **tech-compete** nodes host competitions among alternative technologies on the basis of lifecycle cost (see Chapter 4). In the mining tree, the disaggregation of products and process stages is built from fixed-ratio nodes, and the technology choices — which extraction method, which crusher, which boiler, which class of motor — occur at the tech-compete leaves.

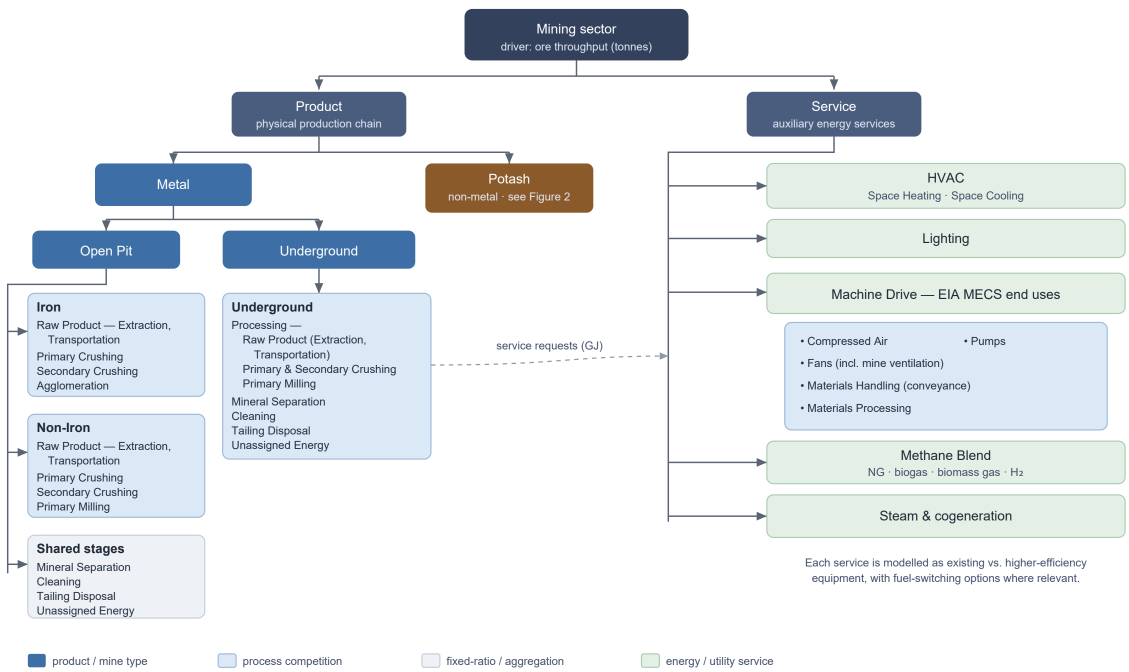

Figure 16.1: Energy-service flow model of the mining sector (region-agnostic master tree). The sector divides into a Product sub-tree — metal (open-pit iron and non-iron, and underground) and potash — and a Service sub-tree of auxiliary energy services structured on the EIA MECS end uses.

16.3.1 Products: metal mining

Metal mining divides first into **Open Pit** and **Underground**, reflecting the substantial difference in their energy profiles (most importantly the ventilation load that underground operations carry). Open-pit mining is then split into an **Iron** path and a **Non-Iron** path, because iron ore and base/precious-metal ores follow different processing routes:

- The **Iron** path runs ore through a Raw Product stage (Extraction and Transportation), Primary and Secondary Crushing, and an **Agglomeration** stage that binds fine iron concentrate into a form suitable for downstream ironmaking.
- The **Non-Iron** path runs ore through Raw Product (Extraction, Transportation), Primary and Secondary Crushing, and **Primary Milling** (fine grinding), but no agglomeration.

Both open-pit paths feed a set of shared stages — **Mineral Separation**, **Cleaning** (concentrate drying), **Tailing Disposal**, and an **Unassigned Energy** node. **Underground** mining carries the same processing chain (grouped under a Processing node: Raw Product, Primary and

Secondary Crushing, Primary Milling) together with its own Mineral Separation, Cleaning, Tailing Disposal, and Unassigned Energy nodes.

The **Unassigned Energy** node is a reconciliation device: it carries metered sector energy that has not yet been allocated to a specific process stage, so that the modelled total matches reported energy use while the more detailed allocation is still being developed.

TODO: Progressively reassign the **Unassigned Energy** at each mine type to specific process stages and technologies, and reduce it toward zero as the bottom-up representation improves.

16.3.2 Products: non-metal mining

Non-metal mining is represented in the same Product sub-tree as a sibling of Metal. The dominant non-metal commodity by energy use is **potash**, whose structure is shown in Figure 16.2. Potash production is met from two distinct mining methods that have markedly different energy profiles, and the model represents both:

- **Conventional mining** — underground extraction (at depths around 1,000 m), with its own Extraction, Transportation, Crush-and-Screen and Dissolution stages.
- **Solution mining** — used for deeper deposits (beyond roughly 1,500 m), where the ore is dissolved in situ and pumped to surface, with a simpler Extraction and Transportation chain.

Both methods feed a shared surface-processing and finishing train — **Crystallization** and **Flotation** (under Processing), followed by **Drying**, **Compaction** and **Tailings Disposal** — and, like the metal mines, both draw on the cross-cutting energy and utility services.

Canada's other non-metal mining is smaller in energy terms but is within the scope of this sector and is included here for completeness. The most significant omissions are uranium and diamonds, both of which are material parts of the Canadian mining industry even though their energy intensity is lower than that of metal milling.

TODO: Add **uranium mining** (Saskatchewan, Athabasca Basin — e.g. McArthur River and Cigar Lake). Saskatchewan supplies roughly a quarter of world uranium; it is currently absent from the sector.

TODO: Add **diamond mining** (Northwest Territories) and other **northern non-metal mining**, once the corresponding regional data are in place.

TODO: Confirm whether lower-energy non-metal extraction such as **salt, gypsum and aggregates** should be represented explicitly or remain aggregated, given their small share of mining energy.

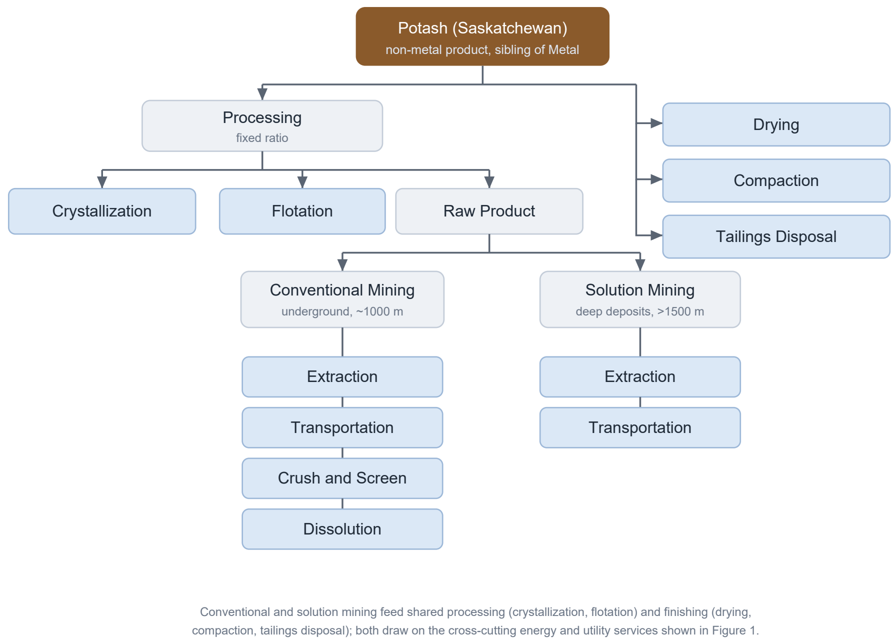

Figure 16.2: Potash flow model. Potash is a non-metal product sibling of Metal; conventional (underground) and solution mining feed a shared processing and finishing train.

16.3.3 Auxiliary energy and utility services

Mining's production technologies draw on the shared **Service** sub-tree rather than specifying their own utility equipment. These services are common to the industrial sectors and are documented in their own chapters in the General Services part; their general structure — and the "existing versus more efficient equipment, with fuel-switching where relevant" convention they follow — is set out in the Sector Overview. This chapter does not repeat that detail. The services the mining sector draws on are:

- **Machine Drive** — see the Industrial Machine Drive Systems chapter. Mining's largest electrical loads sit here: comminution under Materials Processing, mine ventilation under Fans, and haulage and conveyance under Materials Handling, alongside Pumps and Compressed Air.
- **Steam** — see Low-Temperature Industrial Heat, which covers boiler-raised steam and its fuel options.
- **Cogeneration** — see Cogeneration for combined heat-and-power.
- **Methane Blend** — the gaseous-fuel supply node, where natural gas competes with biogas, biomass gasification and hydrogen.
- **HVAC** (space heating and cooling) and **Lighting** for mine facilities.

TODO: The current regional input files still implement an earlier, more detailed auxiliary structure (per-horsepower machine-drive classes; separate pumping, compression, ventilation and conveyance sub-trees) and predate the Product/Service split. Migrate these files to the region-agnostic master tree and the shared General Services chapters referenced above.

16.4 Decarbonisation technologies

Mining's emissions are dominated by diesel in mobile equipment and by the electricity and combustion that drive comminution, process heat, and ventilation. The main decarbonisation levers therefore sit at the **Extraction** and **Transportation** nodes (mobile equipment), the comminution stages (**Crushing**, **Milling**), **Cleaning** (concentrate drying), and underground ventilation, alongside the shared services documented in the General Services chapters. The technologies below are admitted to the relevant tech-compete nodes; representative performance and maturity are summarised in Table 16.1. The figures quoted are indicative ranges from industry and techno-economic reviews (Mungyeko Bisulandu et al. 2026), not the calibrated values used in the model.

Table 16.1: Indicative performance and maturity of key mining decarbonisation technologies

Technology	Indicative benefit	Maturity
Trolley-assist diesel-electric haulage	40–60% energy saving on loaded climbs; ~28–36% CO_2 cut on a low-carbon grid	Commercial
Battery-electric haul truck	Eliminates tailpipe diesel; range / charging constrained	Early commercial
Hydrogen fuel-cell haul truck	Zero tailpipe emissions; H_2 supply constrained	Demonstration
Biodiesel / renewable diesel	Drop-in lifecycle CO_2 reduction, no new equipment	Commercial
High-pressure grinding rolls (HPGR)	~20–30% lower downstream grinding energy	Commercial
Vertical / stirred mills	~40% energy saving vs ball mills; integral drying	Commercial
Ore sorting / pre-concentration	Rejects barren rock; lowers energy per unit metal	Commercial (deposit-specific)
Ventilation on demand (VOD)	Up to ~50% ventilation energy saving	Commercial
Electric / heat-pump drying and boilers	Displaces gas- and oil-fired process heat	See Low-Temp Heat
High-efficiency motors + variable-speed drives	Lower auxiliary drive energy	See Machine Drive

16.4.1 Mobile equipment and haulage

Diesel haulage is the largest single diesel load at most surface mines. **Trolley-assist** systems feed the truck's electric drive from an overhead catenary on the energy-intensive loaded uphill segments, cutting energy use on those segments by roughly 40–60% and lifecycle CO_2 by about 28–36% where the connected grid is low-carbon; the technology is commercial and deployed at several open-pit operations (Mungyeko Bisulandu et al. 2026). **Battery-electric haul trucks**, now offered by the major equipment makers, remove tailpipe diesel entirely but are constrained by range and charging time and remain early-commercial; **hydrogen fuel-cell haul trucks** are at the demonstration stage. **Biodiesel and renewable diesel** are drop-in fuels that cut lifecycle CO_2 without new equipment and are already represented in the tree (the "Extraction Biodiesel" and "Biodiesel truck" technologies). In CIMS these options compete at the Extraction and Transportation nodes.

16.4.2 Comminution efficiency

Comminution is the single largest electricity consumer at a metal mine, so efficiency gains here have an outsized effect. **High-pressure grinding rolls (HPGR)** pre-fracture ore between counter-rotating rolls and reduce downstream grinding energy by about 20–30%. **Vertical roller and stirred mills** can deliver on the order of 40% energy savings relative to conventional ball mills and combine grinding with drying in a single unit. **Ore sorting and pre-concentration** reject barren rock before milling, lowering the energy spent per unit of contained metal, though their applicability is deposit-specific (Mungyeko Bisulandu et al. 2026). These compete at the Crushing and Milling nodes, while ore sorting acts as a service-side reduction in the feed sent to milling.

16.4.3 Electrified and low-carbon process heat

Concentrate drying (Cleaning) and process steam are met largely by combustion today. The low- and zero-emission alternatives — electric and heat-pump drying, electrode and electric-resistance boilers, biomass and hydrogen boilers, and mechanical vapour recompression — are described in the Low-Temperature Industrial Heat chapter and reached through the **Steam** service; the **Methane Blend** node additionally substitutes biogas, biomass gas, and hydrogen for natural gas in any remaining gas-fired duty.

16.4.4 Ventilation and auxiliary drive (underground)

Ventilation can account for roughly 40–50% of an underground mine's energy use. **Ventilation on demand (VOD)** matches airflow to where vehicles and crews are actually working and can cut ventilation energy by up to about 50% (ABB 2024); **high-efficiency fans and motors with**

variable-speed drives further reduce auxiliary drive energy. These are represented through the Fans and other machine-drive end uses described in the Industrial Machine Drive Systems chapter.

16.4.5 Potash-specific options

Conventional potash mining already relies on electric continuous miners and electric hoisting, so its decarbonisation depends mainly on grid emissions intensity and on electrifying surface haulage; VOD applies in the underground workings. Solution mining and the surface **Drying** and **Compaction** stages are heat-intensive and decarbonise through the same low-carbon heat options as above — in particular mechanical vapour recompression for crystallization and evaporation duties, and electric or biomass boilers for drying.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

16.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output (node-unit) basis, a service request for the energy or upstream service it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

A few sector-wide conventions are worth noting. Mining technologies are discounted at a high **financial discount rate** (35% in the current data), reflecting the short payback horizons and high hurdle rates typical of capital allocation in resource industries. **Heterogeneity (variance) parameters** at the process nodes are set so that competitions respond to cost differences without collapsing to winner-take-all. Fuel prices entering the service-cost calculation are scaled by a sector-level block of price multipliers, allowing the mining sector to face prices that differ from the generic delivered price of each fuel.

Greenhouse-gas costs enter through the fuels: combustion emissions follow from each technology's fuel service requests and the prevailing carbon price, so fuel-switching — from diesel haulage to biodiesel or electric, or from natural gas to hydrogen or biogas in the methane blend — is the model's main channel for emissions reduction.

TODO: Mining has no carbon-capture (CCS) service in the current tree (see the Carbon Capture, Utilization, and Storage chapter), on the assumption that its emissions are predominantly combustion-related and dispersed. Confirm this is appropriate, and add process-emissions representation if any mined commodity warrants it.

16.6 Regional coverage and differences

Because the tree is region-agnostic, every province and territory is built from the same master structure, and a region simply carries no production on the branches that do not apply to it. Regional differences therefore show up in *where* production sits rather than in the shape of the tree:

- **Potash** production is concentrated in **Saskatchewan**, which carries essentially all of the potash branch.
- **Iron ore** is concentrated in the **Atlantic** region (the Labrador Trough, on the Québec–Newfoundland border) and **Québec**, which carry most of the open-pit iron path.
- **Base- and precious-metal mining** is spread across **British Columbia, Ontario, Manitoba, Québec** and the **Atlantic** region (for example nickel in Manitoba and Ontario, and copper and gold in British Columbia), populating the non-iron and underground branches.
- **Territorial mining** — diamonds in the Northwest Territories and gold and iron ore in Nunavut — populates the metal and (future) non-metal branches in those regions.

TODO: Several regions are not yet populated in the current input files. Build out mining throughput and technology data for **Alberta** and the three **territories** (the territories are not yet represented anywhere in CIMS — see the Architecture chapter), so that the region-agnostic tree is exercised everywhere it should be.

TODO: The Atlantic, Manitoba, Ontario and Québec metal mines are currently represented generically rather than by commodity. Consider whether the **Iron / Non-Iron** split, plus any commodity-specific energy intensities, adequately capture the real mix in each region, and refine where the averaging is too coarse.

16.7 Limitations and planned extensions

The mining sub-model is a deliberately simplified representation. The TODOs above record the main gaps between this design and both the current input files and the real Canadian mining sector: the auxiliary-service migration to the MECS structure; the reassignment of Unassigned Energy; the addition of uranium, diamonds and the smaller non-metal commodities; full provincial and territorial coverage including Alberta and the territories; and the question of commodity-level detail within metal mining. Together these define the path from the current state of the model to the region-agnostic, fully populated design described in this chapter.

TODO: Replace the illustrative real-world figures and the example discount-rate value in this chapter with quantitative values drawn directly from the model input files, and add a benchmark of modelled mining energy use against CIEEDAC / Natural Resources Canada data.

17 Industrial Minerals

The industrial minerals sub-model of CIMS generally includes all industry involved in the production of all non-metallic mineral products. It encompasses the cement, lime, glass and brick industries, with cement and lime as the major energy consumers. These key products both require significant quantities of thermal energy per unit production during the calcining process. Other energy-demanding processes involved in cement or lime production typically include raw materials preparation (this excludes extraction of the raw materials from their in situ locations, even though these may be within the plant gate), and finishing or finished materials preparation. Calcining remains the most energy intensive part of the processes, consuming large quantities of combustible fuel (85% or more of the total energy consumed). Initial and final preparation of the materials requires electricity to drive motors that are attached to various auxiliary and grinding devices.

Figure: Energy Flow Model for the Chemical Products Industry

18 Chemical Products

Unlike some of the other industrial sectors modelled in CIMS, the chemical products sector is not dominated by a single energy intensive product to which all others can be associated. As a result, several products and processes must be modeled to accurately reflect the energy flow within the industry. The dominant chemical products, from an energy consumption perspective, are: chlorine, caustic soda (sodium hydroxide), sodium chlorate, hydrogen peroxide, ammonia, methanol, ethylene, propylene and polymers. Sub-nodes in the model represent different process stages in which energy consumption can be distinctly represented.

Ethylene, methanol, toluene and xylene are the primary petrochemicals produced. Crude oil and natural gas feedstocks provide the raw material from which these are produced. Ammonia, chlorine (often produced in complement with caustic soda) and sulphuric acid are the major inorganic chemicals produced. Like petrochemicals, natural gas provides the principal raw material used in the manufacture of ammonia. Ammonia, in turn, forms the primary feedstock for urea, ammonium phosphate, ammonium nitrate, ammonium sulfate and nitrogen solutions commonly sold as fertilizers.

Figure: Energy Flow Model for the Chemical Products Industry

19 Pulp and Paper

The pulp and paper industry includes establishments primarily engaged in manufacturing pulp, paper, paperboard and building and insulation board. These establishments are included in the CIMS pulp and paper sub-model because they are large in size and their products are highly energy intensive when compared to other industry products. Paper products, which do not form part of the sub-model include: asphalt roofing, paper box and bag, and other converted paper product industries. These are included in the other manufacturing model of CIMS. The pulp and paper industrial branch has a high energy-intensity per tonne of product.

Pulping processes can be broken down into three basic types: chemical, mechanical, and recycled. Mechanical pulping, which tends to be very electricity intensive, uses wood fibre much more efficiently than does chemical pulping. Recycled pulp consumes considerably less energy than either of the other two major processes but input supply is limited and the collection and transportation of this supply can significantly increase the energy intensity of the product. The products generated from each of these processes are distinct enough that they generally do not compete on the open market.

Each pulp product serves as the stock for many types of paper products. Newsprint, the largest category in terms of production ($> 60\%$), can use feedstock from each of the pulping processes. Other paper products incorporated into this sub-model includes tissue paper, uncoated and coated paper (or wood-free paper), and linerboard, each of which may have varying degrees of stock inputs.

The following steps in the production of pulp and paper are common to both chemical and mechanical mills where raw logs serve as feedstock: wood debarking, wood chipping, pulp making, pulp washing, pulp bleaching, pulp drying, paper stock preparation, paper sheet formation, paper pressing and paper drying. Mills that use chips or recycled paper as feedstock may exclude some of these steps.

Figure: Energy Flow Model for the Chemical Products Industry

20 Metal Smelting and Refining

The metal smelting and refining model of CIMS includes establishments that are primarily engaged in manufacturing finished metal products excluding iron and steel. These processes consume very high amounts of energy per unit of product. The CIMS sub-model explicitly includes processes related to aluminium, nickel, copper, zinc, lead, magnesium and titanium. Though economically important, minor elements such as gold, silver, platinum and cadmium are not represented explicitly because they are often processed in conjunction with the metals listed or are processed in too small a quantity to require direct representation.

In metal smelting, metal is typically extracted from the concentrate by leaching (hydrometallurgical recovery) or through heat (pyrometallurgical recovery). In Ontario, for example, all zinc smelters use hydrometallurgical techniques while copper and nickel smelters use pyrometallurgy. Smelting processes generate products that contain between 95% and 99% pure metal. Refining, if it occurs, depends on the type of smelting technique used. Products prepared through hydrometallurgy tend to use electrowinning for refining while products prepared through pyrometallurgy tend to use electrolysis or fire refining. Refined products are at least 99.99% pure metal.

Purification of the aluminum is accomplished utilizing the Hall-Heroult process, which requires large quantities of electrical energy. Although Canada has no aluminum ore body of sufficient concentration to warrant extraction, readily available, inexpensive electrical energy attracts companies who wish to refine the metal from alumina (concentrated ore). Recycled aluminium requires only 5% of the energy of original production. Increasing demand for conservation and recycling may have a significant effect on the production of virgin metal.

Figure: Energy Flow Model for the Chemical Products Industry

21 Iron and Steel

The CIMS iron and steel model represents this industrial branch as a process whole, including the production of coke from coal, the incorporation of iron ore (including any agglomeration that occurs on site) and ending with the set of end products. End products include the three basic forms (slabs, blooms and billets) and, in many cases, their sub-forms. The iron and steel industrial branch has a high energy intensity per tonne of product.

Presently, two different processes generate Canadian steel. In the first process, coke, a coal derivative, reduces iron oxides in ore to pig iron in a blast furnace. Basic oxygen furnaces (BOFs) then purify this liquid iron along with some scrap by injecting high purity oxygen, which is itself an energy-intense product. In the second process, electric arc furnaces (EAFs) recycle 100% scrap metal. In both cases, the molten steel (with carbon content less than 2%) may be cast, inspected, reheated and finished. Each of a wide variety of finished products requires varying inputs of heat and mechanical energy to its manufacture.

The steel industry is rapidly modernizing. Most of these technologies concentrate on energy-saving measures and reuse of what had historically been waste heat. Direct reduced iron and direct smelted iron processes, presently in pilot scale plants, may eventually make the coke-dominated process redundant. More advanced finishing processes such as direct rolling and thin slab and thin strip casting will eliminate reheating of steel (typically done after initial inspection).

Figure: Energy Flow Model for the Chemical Products Industry

22 Light Industrial

The CIMS industrial sub-model for other manufacturing industries has been created to capture those industries within a region, which, on their own, do not consume enough energy to merit the development of a separate model. When considered as a group, however, they may consume a significant (if not the dominant) portion of the energy within any one region. The industries considered part of the other manufacturing sector are: food, beverage, tobacco, rubber products, plastic products, leather and allied, primary textiles, textile products, clothing, wood, furniture and fixture, printing and publishing, fabricated metal products, machinery, transportation equipment, electrical and electronic, and other manufacturing.

The other manufacturing group of industries typically includes a large variety of technologies and processes. Usually, a simplified flow model of generic energy services represents adequately the energy consumption of these industries as a whole. Steam boilers, process heat, space heat, and electricity for lights and electric motors and their attached auxiliary devices (pumps, conveyors, fans and the like) constitute the bulk of energy-using services.

Unlike other "major" industries the other manufacturing industry as a group does not have any single product which dominates the production processes and to which the production of all other products may be linked. As a proxy for the physical output measures used in the sub-models major industries (i.e., tonnes of steel or cubic metres of gasoline) the output of the other manufacturing industry is represented in monetary terms such as the RDP (real domestic product) of the industry.

Figure: Energy Flow Model for the Chemical Products Industry

23 Construction

TODO: Add a description of the Construction sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

24 Residential

The CIMS residential sector sub-model attempts to capture all energy demand by residential buildings within Canada. This includes single family dwellings, duplexes, row housing, walk-ups and large apartments. Large apartment buildings are often considered commercial buildings by utilities, but, since their energy consumption patterns better fit within the residential sector, they have been included as part of this sub-model.

The number of households drives the residential sector models and is calculated exogenously, based on future population growth estimates. This variable directly determines the demands for all services, except for water heating and furnace fan demand. The technologies for services that require hot water and furnace fans establish those demands.

24.1 Energy Use in the Residential Sector

In 2002, the residential sector accounted for approximately 17% of all energy consumed in Canada. This energy provides services such as space heating, water heating, lighting, cooking, refrigeration and clothes drying. Space heating accounts for about 60% of residential energy consumption, water heating approximately 21% with the remainder going to lighting, appliances and other uses. The exact nature (i.e., quantity, fuel type etc.) of this consumption depends on many variables such as availability, climate, building type and occupancy patterns.

The following describes energy end-use services common to residence types incorporated into the model, the variables that affect the amount of energy consumed in providing the services, the relationship between the end-use and the driving variable at the primary node (number of households) and any other interrelationships that permit a more accurate simulation of the end-uses.

Space Heating

A variety of factors uniquely determine the energy consumption required to heat a household. These factors include the physical building envelope, the type of heating system, maintenance of both the envelope and heating system, climate and exterior landscape features, and thermostat settings. Modelling this service requires many assumptions and generalizations. Due to the multitude of housing types, the sector must be condensed into groups consisting of thousands of households which are represented in the model by one typical house archetype. All heating systems cannot be included, due to the large number of possibilities; thus the models contain only the most common systems and those considered likely to gain significant market shares in the future. Case studies and surveys provided estimates for geographical and behavioural assumptions.

Each space heating technology in the residential CIMS sub-model combines a shell archetype and a heating system. Each archetype, which is a set of physical measurements for components such as floor space, window area, insulation levels, and rate of air exchange, represents a group of households. Modellers use archetype descriptions to calculate the heatload and cost of the typical building as well as the cost and energy savings of retrofitting. For each shell archetype, CIMS includes a choice of four or five different heating systems. These systems include oil furnaces, natural gas furnaces (typically three efficiency levels), high efficiency space heat/hot water systems, wood stoves, electricbaseboard heating, and electric air source heat pumps.

Non-Appliance Hot Water Use

This service meets the hot water required for baths, showers, hand washing, and cleaning. The competing technologies, shower heads and faucets with varying flow (litres/minute) levels, consume hot water rather than energy. The water demand determined by this node contributes to the hot water demand that drives the hot water heating service.

Lighting and Major Appliances

Each of the following appliances is modelled as a separate service:

- Refrigerators
- Freezers
- Dryers
- Ranges
- Dish Washers
- Clothes Washers
- Air Conditioners
- Lighting

The stock is measured in number of appliance units (i.e., number of refrigerators) and the ratio between the service and the driving variable (number of households) is the market penetration of the appliance. Generally, there is a range of efficiency levels for new stock in each major appliance service.

Clothes washers demand both electricity and hot water. Dish washers are disaggregated into two categories: machine, which demand electricity and hot water, and non-machine, which only consume hot water. Due to varying demand for air conditioning not all the residential sub-models incorporate them as a separate end-use and they are often combined with the minor appliances.

Minor Appliances

This service contains many diverse technologies including televisions, car block heaters, lawn mowers, and computers. Although, these appliances collectively consume a significant portion of household energy, individually either their consumption or their market penetration is low. As a group, this end-use is growing considerably. Efficiency improvements are outweighed by increase in number of goods.

Renewable Energy Sources and Other Distributed Energy

A number of renewable technologies (in addition to wood stoves) can be applied at the household level to provide both passive and active energy. These include photovoltaic panels, geothermal, solar hot water heating and passive solar design. Options are also increasingly becoming available to generate electricity using combined heat and power generations at the household level, for instance through fuel cells and through micro-CHP (for instance the stirling engine). Some of these options are modelled at the end-use in which they meet thermal demand, rather than as a unique 'supply' node in the flow model.

Water Heating

The water heating demand depends on the technologies used to supply services of non-appliance hot water, dish washing, and clothes washing. For most purposes, hot water must be at 60oC and, unheated water generally enters the household at 15oC. Both storage tank and non-storage tank water heaters are modelled. Fuel cell and stirling engines are modelled at this end-use to meet water heat loads (while generating electricity for household use).

24.2 Energy Flow Model, Residential Sector

Although each province in Canada has a number of unique characteristics in terms of climate, the availability of fuel and the structure of the residential sector, all households require the same basic set of energy services. The following section describes the basic structure and the end-uses common to all regional residential models. Major differences between regional models are described later in this section.

Figure 5 presents CIMS's residential sector flow sub-model. The number of residences required in any year drives the model. The flow model demonstrates the linkages between the driving variable and the energy end-uses in the residential sector. The number of residences is determined exogenously, based on future growth estimates of population and economic activity. The end-uses are linked to the driving variable through engineering ratios based on historic relationships between a residence and each end-use.

Energy End-Uses (Non-Space Heating)

Due to similarities in service demand and technology choices, the lighting and major appliance services modelled in the residential sector have been aggregated for all housing types. Air conditioning, a large energy consumer in some regions (e.g., Ontario), consists of room or window units and central units. Disaggregating apartment and non-apartment water heating captures the level of use and technology differences between these two categories.

Space Heating

In contrast to the other energy services in the residential sector, the amount of space heating demanded by a residential unit can vary considerably depending on the building type. As a result, the space heating service node of the model is divided into three basic housing types: apartment, attached (row), and single family detached dwellings. The apartment category includes both high and low rise apartments while the attached category consists of duplexes, triplexes, and townhouses. As described above each of these housing types have been divided up into different archetypes to describe the differences in building heatloads and the fuels and efficiencies of the heating systems.

24.3 Apartments

In the apartment node, two basic technology archetypes are available, standard and 'improved shell'. The standard archetype generally represents the existing or a baseline shell design that demands a relatively high heat load while the improved apartment shell design represents better insulation, windows, etc., demanding a lower heat load. Each of these archetypes is divided into several different heating system types based on the type of fuel and efficiency of the heating system, e.g., oil heating, several efficiency levels of natural gas heat (62%, 78%, 90%) and electric heat. During each model run new apartments can be purchased either from the options available from existing stock or from a selection of new buildings with an improved shell design and better heating systems. There are no retrofit options currently available for existing apartment stock.

24.4 Single Family Dwellings

The greatest level of technological variety in the residential sector exists in the types of systems for heating single family dwellings (SFDs). These technologies also show the greatest differences between the regions. In the basic CIMS residential sub model, SFDs are divided up into two categories, existing and new housing.

Two archetypes are used to model all existing houses. Archetype A houses have very little insulation and, therefore, higher heatloads than Archetype B houses. Where wood consumption is present, wood stoves meet part of heat load demand. CIMS allows houses in either of these archetypes to retrofit to improve insulation levels. The six possible heating systems for the current housing stock are oil furnace, electric baseboard heating, air-source heatpump, standard natural gas furnace (78% efficient), high-efficiency natural gas furnace (78% efficient), and high efficiency space heat/hot water systems.

In each region, three types of houses compete for new stock: standard, improved and R-2000 archetypes. R-2000 houses meet the current heating budget specified by R-2000 standards for the region. The improved archetype is based on the ratio between standard and improved archetypes. Houses can be upgraded to better insulated categories through retrofits. Heating systems are the same as those modelled for existing homes. In some regions wood stoves may be chosen to heat standard and improved archetypes.

24.5 Attached Housing (Row)

The attached housing category is modelled in the same manner as single family dwellings.

Figure 5. Energy Flow Model for the Residential Sector

24.6 Climate Zones

The British Columbia residential sub-model has some differences compared to the other provinces in Canada. It is the only model to make use of the primary/secondary climate distinction available for single family detached homes - coastal SFD (Lower Mainland and Vancouver Island) and interior SFD are represented. The same has not been done for apartment and attached housing since over 80% of these buildings are in the coastal area and this trend is assumed to continue over the modelling period.

The coastal SFD (Primary Climate) housing portion of the model is as described above. The interior region (Secondary Climate), with only half as many houses as the coastal region, uses only a single archetype to represent existing stock. The archetype heatload is the average of all current housing stock heatloads. These houses, like the others, can be retrofitted to improve insulation levels. Since wood provides primary heat for approximately 20% of interior houses, CIMS includes wood stove heating systems for these archetypes. The other heating systems in the interior are identical to the coastal systems described above.

25 Commercial and Institutional

25.1 Energy Use in the Commercial Sector

In CIMS, the definition of the commercial sector excludes light industrial facilities, which are included in the other manufacturing sub-model, and high-rise residential apartments, which are included in the residential sub-model.

The commercial building sector accounts for roughly 20% of all energy consumed in Canada. For the most part, energy in the commercial sector is consumed to provide energy services such as lighting, heating, ventilation and air-conditioning (HVAC); cooking, refrigeration, hot water, and plug load (i.e. power for computers and other office equipment). Lighting and HVAC account for roughly 60% of the energy consumed in this sector. However, the distribution of energy use in any particular commercial building depends on its architectural characteristics, building type and size, efficiency of installed equipment and activities of the occupants.

The most common fuels used in the commercial building sector are, in order of importance, electricity, natural gas and light fuel oil. Electricity is the only energy source used for lighting, refrigeration, plug load, cooling, fan and pumping. Electricity and natural gas are used for cooking. All three energy sources are used for space and hot water heating.

25.2 Energy Flow Model, Commercial Sector

Figure 6 shows CIMS's commercial sector energy flow model. CIMS considers commercial floor space as being the driving variable (primary node) of the commercial sector. The flow model demonstrates the linkages between the driving variable (commercial floor space) and the energy end-uses in the commercial building sector. The amount of commercial floor space is determined exogenously, based on future growth estimates of population and economic activity. The end-uses are linked to the driving variable through engineering ratios based on historic relationships between commercial floor space and each end-use. Not all energy end-uses found in the commercial sector are included in the model. Only space heating, lighting, refrigeration, cooking, hot water and plug load are considered significant enough, in terms of their total energy consumption, for inclusion.

End-uses are described in more detail in the next sections.

Figure 6. Energy Flow Model for the Commercial Sector

Lighting

In the commercial building sector, lighting consumes more electricity than any other end-use, accounting for 30 to 40 percent of total consumption. Lighting is measured in terms of the number of square meters of commercial floor space that it serves. Therefore, the lighting service has a one to one ratio with commercial floor space.

Lighting is not modelled separately by building segment. CIMS simulates lighting at different energy efficiency levels. The base technology represents the average efficiency of stock in 2000. The 'new' technology represents the marginal energy efficiency of new stock built in 2000. Other options represent actions to adopt more efficient lighting equipment (super T8, advanced HID, compact fluorescent lighting) and improving lighting design.

The lighting end-use category is modelled as three sub-categories: general area lighting, high bay lighting and service lighting. General area lighting is the dominant category of lighting and refers to the lighting found throughout general office areas and uses predominately fluorescent tube lights. Service area lighting refers to lighting systems used in corridors, lobbies, vestibules, washrooms, and any non-critical areas adjacent to areas illuminated by general area lighting. Lastly, service lighting refers to the lighting systems used in high bay areas of a building - large atria and lobbies, gymnasiums etc. The energy intensity of any sub-category depends on the efficiency of the lighting equipment and the use rate.

CIMS does not currently account for the energy consumed for exterior lighting such as street lighting, architectural lighting or parking lot lighting.

Shell/HVAC

The Shell/HVAC category is disaggregated into nine representative building segments (sub-categories). These segments, listed below, combine two or three different building types to group buildings with similar HVAC system types and usage. For example, the schools category includes elementary schools, colleges and universities.

- Warehouses and Wholesale Outlets
- Hotels and Motels
- Schools and Universities
- Small Office Buildings
- Large Office Buildings
- Small Retail
- Large Retail
- Hospitals and Nursing Homes
- Miscellaneous Buildings (not including residential or light industrial)

The mix of buildings in the segments is derived from data for the base year. The eight segments each have different occupancy and use rates, system types, and shell types. The relative shares of the building types within each segment can vary over the study period.

Each building segment encompasses a variety of shell and building HVAC technology options. Each is represented by a 'shell' node which competes alternative shell options. This represents a demand for heat load which is met by a corresponding HVAC service node for each building segment (see flow model). The link between the shell and HVAC competitions attempts to capture the relationship between decisions about building shell characteristics and the amount of energy demanded of the heating and air conditioning services. HVAC systems are modelled as 'single' technologies to represent the interdependence of choices about heating, cooling and ventilation systems.

Refrigeration, Cooking, Hot Water, Plug Load

The end-uses in the commercial sub-model are linked to the driving variable based on historic ratios of use per unit floor space and the weighted average of all the various commercial sub-sectors (e.g., a hospital will use more hot water than a warehouse per unit area). The weighted ratios can change over time, depending on changes in the proportion of the different kinds of floor space assumed over that period.

Domestic hot water (DHW) end-use represents the hot water used for all building services except space heating and is measured in cubic meters of hot water demand for the whole commercial building sector. It is related to the primary node by the ratio of cubic meters demand to commercial building floor space. Hot water is used for showering, washing, cleaning and food preparation.

The refrigeration end-use includes all commercial refrigeration except for cooling required for air conditioning and is measured in gigajoules of refrigeration demand per square meter of commercial building floor space. This end-use is represented by two types of technologies; stand-alone plug-in refrigerators and large built-up refrigeration systems. These two types have different efficiency improvement opportunities. The refrigeration category in the model combines these two technologies and its energy intensity is the weighted average of the two system types. The ratio of the two system types is assumed to remain constant over time in a typical simulation.

The cooking end-use is served by either electric or gas ovens, stoves and fryers and is measured in gigajoules of cooking demand per square meter of commercial floor space. This end-use also consists of different system types and the energy intensity of this end-use is the average for all types of commercial cooking equipment. As in the other end-use services, the shares of the various equipment types are assumed to remain constant over the simulation period.

25.3 Assumptions and Caveats

Any model is a simplification of reality. The results from CIMS are limited by data availability and the complexity of modelling all energy relationships in the commercial sector. The major assumptions and caveats made in the process of developing this CIMS sub-model are described below.

25.4 Energy Prices

As with the other CIMS sub-models, energy prices used in the commercial sector sub-model for both natural gas and electricity are based on the average charge per unit of energy consumed. Each unit of energy is assumed to cost the same regardless of the consumption rate of the customer. In reality both natural gas and electric utilities have rate structures that charge different prices depending on a customer's total consumption. Therefore energy cost savings predicted by CIMS may be different from actual savings.

Many utilities also have demand charges or fixed fees charged per kW of demand. Since many energy efficiency measures reduce the demand of a facility, they also reduce the demand charges. These cost savings have not been included in the savings predicted by CIMS.

25.5 Discount Rates

The main criterion used by CIMS for choosing technologies is the lifecycle cost of competing technologies. To reflect the constraints imposed on investments due to rapid payback requirements, risk aversion, capital availability and other investment criteria, annualized costs in the commercial sector are generally derived using a 40% discount rate. Cogeneration technologies, however, are discounted at a rate of 30% because investors in cogeneration are generally seen to have a longer planning / finance horizon than investors in energy efficiency. In addition, the hospital and school sectors' shell/HVAC technologies are discounted at a rate of 30%, instead of 40%, consistent with evidence of their longer planning / cost horizons and the fact that they are owner-operated. These rates reflect ex post measures of discount rates and are meant to encompass many more factors than simple rate-of-return criterion.

25.6 Retrofit Technologies

Some technologies are also set up as retrofits in the commercial sector. After existing stock is retired, CIMS compares the costs of the parent or lead technology and the attachable retrofit technologies. Although it remembers how much of the original parent stock there was, at this point CIMS assumes that there is no existing stock and sets up a competition between the lead technology and the retrofit technology(ies) to determine how much of that ”remembered” original stock will remain as is (no retrofit) and how much will be retrofitted. The cost of the parent technology is simply the operating and maintenance (O&M) cost including fuel costs. The cost of each retrofit technology is O&M, fuel plus the annualized retrofit capital cost less any tax or other credit.

25.7 Interactive Effects Between Various Energy End-uses

The commercial building sector is different from both the industrial and the residential sectors in that there are significant interactive effects between energy end-uses and energy efficiency measures. As a result of these interactive effects, an energy efficiency measure that is undertaken for one end-use may have an impact on the energy consumption of another end-use. For instance, an energy-efficient lighting retrofit in a commercial building will reduce the cooling load but will increase the heating load because the waste heat from the lighting system is reduced. However, because there are so many factors that influence the consumption of energy by the HVAC system, these interactive effects are very difficult to quantify. Models exist that calculate these interactions for a specific building type but this type of analysis is beyond the scope of this current CIMS sub-model. The focus in the model is on capturing the main interactive effects between shell choice and HVAC demand.

25.8 Cogeneration

Cogeneration of heat and electricity has been limited exogenously in CIMS to between 30 and 50 percent of the demand in schools, large office, hospital, miscellaneous and hotel shell/HVAC categories. This is because cogeneration is only practical in large facilities that have a demand for both electricity and hot water or steam. The range of 30 to 50 percent is due to the different mix of buildings that make up each sub-category.

Cogeneration systems are sized to provide 100% of the facility’s heating load. In certain instances this results in a surplus of electricity beyond that required by the shell/HVAC system. This surplus is assumed to be sold back to the electric utility for the same price that the facility is charged for electricity.

25.9 Oil Heating

CIMS assumes that no oil-fired heating is installed after 1990 in the commercial building sector of all regions except the Atlantic region, where natural gas is less available. This was done to simplify the model since oil heating is a small and declining portion of the market. The

Atlantic region is the only region in Canada where the proportion of oil-fired heating is significant in the commercial building sector.

25.10 Lighting

CIMS does not account for the energy consumed for exterior lighting such as street lighting, architectural lighting or parking lot lighting. Data for these end-uses are not available.

26 Transportation

This document describes the structure and data inputs of CIMS-Transportation (CIMS-T), the transportation module of CIMS. It assumes that the reader is already familiar with the CIMS simulation algorithm and parameters. Additional documentation is available from the Energy and Materials Research Group that describes the general function of CIMS.

TODO: add sections on SAF, e-fuels

26.1 Service Flow Model

Figure 1 presents a simplified version of the service flow model of CIMS-T. The three main energy service categories are personal mobility, freight transport and off-road vehicles. In urban areas, single occupant vehicles (SOV), high occupant vehicles (HOV), public transit, walking and cycling provide personal mobility. Personal vehicles, buses, planes and trains provide transportation between urban areas. Marine, air, road and rail modes provide freight transportation services.

Figure 5 Service Flow Model of CIMS-Transportation

The SOV, HOV and personal vehicle nodes attached to the urban and inter-city nodes in the service flow model represent a demand for passenger vehicle services. The passenger vehicle service node located to the bottom left of the diagram supplies this demand. Personal vehicles are designated as either cars or trucks in CIMS. Within these categories, vehicles of different efficiency levels and fuel types (gasoline, diesel, natural gas, propane, methanol, ethanol, battery electric, gasoline-electric hybrid, hydrogen fuel cell) compete to meet demand. Efficiency and fuel type are specified only for new passenger vehicles. The nodes representing old and recent passenger vehicles are used simply to keep track of vintages of vehicles already present in the base year (2000). These vehicles are no longer available to supply new demand for passenger vehicle services. During the simulation, a retirement function removes these vehicles from the existing stock.

26.2 Endogenous Actions

CIMS-T has the capability to simulate the following actions endogenously:

- Changes in the efficiency of the gasoline personal vehicle fleet,
- Fuel switching within the personal vehicle market,
- Shifts between the personal vehicle and public transit modes,
- Shifts between SOV use and HOV use,
- Changes in the efficiency of the freight truck fleet, and
- Changes in demand for personal mobility.

With the exception of efficiency choices between road freight vehicles, technology shares within the freight sector continue to be simulated exogenously. The same is true for the off-road sector. Choices between public transit technologies (bus versus subway) are also specified outside the model. Actions that involve changes in the operation and maintenance characteristics of existing technology stocks are incorporated exogenously as the CIMS models simulate technological change through stock turnover, not through changes in the use and operating characteristics of existing stocks.

26.3 Input Data

The CIMS models are data-intensive: in order to carry out a simulation, input data are required to describe a series of characteristics for each technology. Technology characteristics and data sources for CIMS-T are listed in Table 1. The input data used to describe personal vehicles is described in more detail below, given the importance of this vehicle category to the transportation sector as a whole.

Table 1: Technology Specific Input Data for CIMS-T

Personal vehicles are defined by efficiency and type of fuel / technology. Technological characteristics of alternative fuelled vehicles were initially based on the output of the Transportation Issue Table. Vehicle cost differentials relative to conventional gasoline were taken from Levelton Engineering Ltd.'s work for the Issue Table (1999), with the exception of gasoline-electric hybrid and hydrogen fuel cell vehicles – for these vehicle types we used data from the NEMS model of the U.S. Department of Energy (1998). Energy consumption relative to conventional gasoline vehicles was also estimated based on the Levelton (1999) report. In 2004, a comprehensive literature review was undertaken to update the capital cost and energy consumption values for personal vehicles. Sources such as Azar, Lindgren and Andersson (2003); Ogden, Williams and Larson (2004); and Weiss et al. (2003 and 2000) were used to verify and adjust the existing data points.

The values of the intangible cost parameters within CIMS-T reflect differences between various efficiency levels and fuel types of SOVs, as well as between SOVs and other modes of transportation. The estimates of intangible costs for more fuel efficient vehicles are based on market studies of the revealed willingness-to-pay of many consumers for higher horsepower vehicles, which are generally less efficient (U.S. Department of Energy, 1998; AMG, 1999).

The estimates of intangible costs for alternative fuelled vehicles represent consumer preferences for longer range and wider fuel availability, as well as a general preference for conventional gasoline vehicles. The range and fuel availability parameters were derived by combining technology data of the U.S. Department of Energy (1998) with the results of a stated preference survey carried out by Bunch et al. (1993). The stated preference approach is necessary in dealing with alternative fuelled vehicles because these tend to be emerging technologies for which revealed preference data do not exist.

The estimates of intangible costs for public transit and HOVs relative to gasoline-driven SOVs are based on the financial costs and market shares of alternative modes of urban travel in Canada, as well as comparisons to various European countries where the financial costs associated with personal vehicle travel are much higher.

We are currently reviewing the intangible cost parameters used in CIMS-T based on data from a series of stated preference surveys carried out by the Energy and Materials Research Group and CIEEDAC.

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27 Freight Transportation

TODO: Add a description of the Freight Transportation sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a dedicated written description. General CIMS-Transportation structure, service flow model, and input data are described in the Transportation chapter.

28 Waste

TODO: Add a description of the Waste sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

29 Agriculture

TODO: Add a description of the Agriculture sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

30 Forestry

TODO: Add a description of the Forestry sector.

Placeholder chapter — this sector is represented in the model architecture but does not yet have a written description.

31 Carbon Dioxide Removal / Direct Air Capture

- Run year as normal with carbon price
- Re-run year with shadow carbon price, iterate until between:
  - total emissions difference between runs is equal to maximum DAC quantity
    * have maximum annual growth rate for DAC?
    * Expert opinion?
  - minimum shadow price equals DAC cost
- Emissions difference between runs is DAC request by tech
  - Only change shadow price in sectors that are allowed to purchase DAC
- Total emissions difference is DAC sequestered

32 LULUCF

Need to include emissions for accounting. Want to have total emissions from CIMS equal the NIR for easy comparison.

Also include credits in GHG accounting, again for easy comparison. Or should this be done outside CIMS?

Part IV

General Services

33 Industrial Machine Drive Systems

TODO: Describe how the model represents industrial machine drive systems and their contribution to industrial electricity demand across manufacturing and resource-processing sectors. The section should explain the treatment of end uses such as motors, pumps, compressors, fans, conveyors, and other mechanically driven equipment, including how energy consumption is allocated across industries consistent with sources such as the U.S. Energy Information Administration Manufacturing Energy Consumption Survey (MECS). It should document assumptions related to motor efficiency improvements, variable speed drive adoption, equipment turnover, process electrification, and production-output linkages. The section should also explain how machine drive demand responds to changes in industrial activity, efficiency policies, and technology adoption, as well as any interactions with waste heat recovery, onsite generation, or demand-side flexibility measures.

Nodes shaded in green represent unique industry processes, while nodes shaded in other colors represent auxiliary systems. These are generic energy services that are supplied to the major process technologies and are shared by all CIMS industry sub-models. The auxiliary systems fall into four general categories: steam generation (boilers and cogenerators), lighting, heating, ventilation and air conditioning (HVAC), and electric motor systems including pumps, fans, compressors and conveyors. Figure 9 shows the energy flow diagrams for auxiliary systems not included in Figure 8. In some cases, the energy service meets the direct need for steam, pumping or compression, while in other cases, it serves only to provide suitable conditions for production to continue, as in the case of lighting and HVAC systems.

Figure 9. Auxiliary Flow Model Diagram

The sections that follow describe how key auxiliary systems and process specific components are captured by the CIMS industrial model. The section on auxiliary systems covers steam generation and electric auxiliary services. Lighting and HVAC are not described in detail because the energy demanded by these services is small. The section on process specific systems covers each of the CIMS industrial sub-models.

33.0.1 Auxilliary Systems

33.0.1.1 Steam Generation

A number of industry process technologies generate steam as a by-product. CIMS accounts for steam generated by these technologies in the overall steam demand in each industry. Any shortage of steam must be met by boilers and cogenerators.

Boiler efficiency can vary greatly depending on boiler design, age, and fuel used. For modern oil and gas boilers, thermal (first law) efficiencies may be 85% or higher. Boiler efficiencies can be increased by introducing non-condensing and condensing heat recovery systems and by installing regenerative burners with computerized fuel/air mixtures to maximize fuel efficiency.

Cogeneration is the sequential production of electricity and useful thermal energy, usually as steam or shaft drive power. In cogeneration a boiler is used in conjunction with a turbine system to generate electricity, with the "waste" steam going to meet process steam requirements. In comparison to conventional electricity generation, cogeneration facilities are capable of conversion efficiencies of up to 85% (including both the steam and electricity produced) as compared to utility condensing turbine systems which have an efficiency in the 30% to 40% range.

33.0.1.2 Electric Auxiliary Systems

The vast majority of electricity consumed by industry is used by motor systems. A motor is the core component of a much broader system of electrical and mechanical equipment that provides services, including hydraulic power, compressed air, motive power and air flow. Opportunities for efficiency improvement exist in both the motor itself, and in the latter systems – pumping, air displacement, compression, conveyance as well as other types of machine drive that are unique to a given production process (direct drive).

Figure 3.3 above is a generic energy flow model representing motor and steam auxiliary systems as they appear in the various CIMS industrial sub-models. As the figure indicates, CIMS simulates six auxiliary groups related to process drive (in red). The first five demand shaft or mechanical drive provided by the sixth group, motors (machine drive). Each group simulates technology evolution of a specific system.

Increased disaggregation in some of the groups reflects differing end-uses and levels of efficiency. For example, compressor systems vary in electrical efficiency by size. Therefore, competition between different technologies must be separated by size. The energy flow model reflects this by displaying two sizes of compressors, which are available for use by the model.

Pumping, air displacement, compressing and conveyance systems have been represented as packages within CIMS. The components of each of these systems are associated with the key end-use. For example, a pipe, throttle and speed control device is attached to a pump, and the entire package is called a pump system. In the model, there are at least two types of systems:

1. The least efficient system comprised of the least efficient components
2. The most efficient system comprised of the most efficient components

The system efficiency is then a function of the efficiency of all the components within that system. In some cases intermediate efficiencies have also been listed. New technologies may be added to reflect the intermediate range.

Electronic variable speed drives (VSDs) provide speed control by varying the rotating speed of the induction motor to match variable speed requirements of driven equipment. Pumps and fans, for example, are often required to deliver varying qualities of fluid flow. When this control procedure is used the energy consumption of the fan or pump is dramatically reduced during part load operation, as compared to the conventional practice of using a valve or damper to reduce flow.

34 Low-Temperature Industrial Heat

34.1 Scope and Definition

Low-temperature industrial process heat in CIMS is modeled as the steam, hot-water, and direct-heating services supplied at end-use temperatures below approximately 200 °C. Process heat is partitioned into three temperature bands — low (below 200 °C), medium (200–400 °C), and high (above 400 °C). The bands are defined by end-use temperature rather than by energy or equipment type, because the technology set admitted to the relevant service nodes differs substantially across them. The 200 °C boundary follows the convention adopted by the U.S. Department of Energy, Natural Resources Canada, and applied in the recent peer-reviewed techno-economic literature (e.g., Schoeneberger et al. 2020). Industrial heat globally accounts for approximately two-thirds of final industrial energy use and roughly one-fifth of energy-related CO_2 emissions; analyses across IEA member countries place 30–45 % of industrial heat demand within the low-temperature band (International Energy Agency 2022; International Energy Agency 2024; Madeddu et al. 2020).

The allocation of base-year low-temperature heat demand across sub-models on a provincial basis is the result of a calibration exercise bringing together information from Statistics Canada and Natural Resources Canada's Comprehensive Energy Use Database (CEUD), Environment and Climate Change Canada's National Inventory Report (NIR), and U.S. Energy Information Administration Manufacturing Energy Consumption Survey (MECS) data used for analogous subsector benchmarks. The CEUD does not publish process heat disaggregated by temperature band; end-use accounting nevertheless indicates that conventional steam systems, boilers, and process heating account for the majority of fossil-fuel consumption in Canadian manufacturing (Natural Resources Canada 2024).

34.2 Industrial Subsectors and End-Uses

Low-temperature heat demand is concentrated in a defined set of CIMS industrial sub-models — Pulp and Paper, Chemicals, Mining, Industrial Minerals, Metal Smelting, Iron and Steel, and Light Industrial (which aggregates food and beverage, textiles, wood and prefabricated building products, and general light manufacturing). Each sub-model is calibrated independently for base-year energy consumption, and service demand is assumed to scale with industrial sectoral output projections in future years. The relative share of low-, medium-, and high-temperature heat within each subsector is assumed to remain constant in projection years. Subsectors and representative end-use temperature ranges relevant to the Canadian industrial mix are summarized in Table 34.1; note that food and beverage processing, textiles and laundry services, wood products, and light manufacturing and assembly are not standalone CIMS sub-models but are represented within the Light Industrial sub-model.

Table 34.1: Representative low-temperature end-uses and temperature ranges by industrial subsector

Subsector	Representative end-uses	Range (°C)
Pulp and paper	Black-liquor evaporation, paper drying on steam-heated cylinders, digester heating	100–180
Chemicals and pharmaceuticals	Distillation reboilers for low-boiling solvents, batch reactor heating, crystallization	80–180
Mining and oil and gas	Produced-water heating, glycol regeneration, remote-site building heat, mid-stream gas processing	60–200
Food and beverage processing	Pasteurization, sterilization, spray and tray drying, clean-in-place wash water	60–200
Textiles, leather, and laundry services	Dyeing baths, drying, finishing	60–160
Wood products and prefabricated building products	Lumber kiln drying, panel pressing	60–200
Light manufacturing and assembly	Paint cure, parts washing, fabrication-hall space conditioning	50–180

34.3 Technologies for Low- and Zero-Emission Heat Supply

This section describes the low- and zero-emission supply technologies admitted to the low-temperature service nodes in the CIMS industrial sub-models, together with their assumed performance parameters and data sources. Technology efficiencies and coefficients of performance (COPs) used to parameterize the technology competitions are summarized in Table 34.2. Combustion boiler efficiencies are estimated based on data from MECS; heat pump COPs are based on inputs from IEA Heat Pumping Technologies Programme Annex 58 (U.S. Energy Information Administration 2021; IEA Heat Pumping Technologies Programme 2023).

Table 34.2: Assumed efficiencies and coefficients of performance for low-temperature heat supply technologies

Technology	Efficiency / COP	Maximum temperature
Natural gas boiler	80–85 %	250 °C
Liquid fuel boiler	80–85 %	250 °C
Coal boiler	85–90 %	250 °C
Biomass boiler	75 %	250 °C
Electric resistance / electrode boiler	99 %	250 °C
Heat pump	COP 4.0–6.0	$\leq$ 100 °C
Heat pump	COP 2.5–4.0	100–150 °C
Heat pump	COP 2.0–3.0	150–200 °C (demonstration)
Mechanical vapour recompression	COP 5–10	Evaporator duty only
Solar thermal, flat-plate / evacuated-tube	n/a	100 °C
Solar thermal, linear concentrating	n/a	200 °C

Combustion boilers and electric resistance boilers are assumed to have a lifetime of 25 years; industrial heat pumps and mechanical vapour recompression units a lifetime of 15 years; and solar thermal collectors a lifetime of 20 years.

34.3.1 Industrial Heat Pumps

Closed-cycle vapour-compression heat pumps upgrade ambient or recovered low-grade heat to a useful end-use temperature. Commercially available products reach temperatures of approximately 165 °C; units operating at 200 °C with novel working fluids are at the demonstration stage (Arpagaus et al. 2018; Zühlsdorf et al. 2019). Mechanical vapour recompression, an open-cycle heat pump configuration applied to evaporators in dairy, sugar, and pulp operations, is the most mature variant and is represented as a distinct technology in the food and beverage and pulp and paper sub-models.

34.3.2 Direct Electrification

Electric resistance and electrode boilers deliver low-pressure steam at near-100% end-use efficiency at lower capital cost than heat pumps, and are represented as the simplest like-for-like replacement for packaged gas-fired boilers. Application is favoured in retrofit configurations where capital constraints favour low first-cost equipment (Environment and Climate Change Canada 2024). Direct-contact electric processes (induction, infrared, microwave, and dielectric heating) are commercially deployed for specific drying, curing, and surface-heating duties in the corresponding service nodes (Madeddu et al. 2020).

34.3.3 Solar Thermal

Flat-plate and evacuated-tube collectors economically supply heat up to approximately 100 °C; linear concentrating systems (parabolic-trough and linear-Fresnel) extend the addressable range to 200 °C and beyond. Schoeneberger et al. (2020) estimate that approximately half of global industrial heat demand falls within the temperature window addressable by current commercial solar thermal technologies. Within Canada, solar resources are most favourable in the Prairie region and the southern interior of British Columbia; representation in CIMS uses capacity factors calibrated to regional direct normal irradiance (International Energy Agency 2024).

34.3.4 Bioenergy

Biomass-fired boilers, anaerobic-digestion biogas, and renewable natural gas from landfill and agricultural sources are admitted as drop-in low-carbon supply technologies in facilities configured around combustion equipment. Biomass is the dominant on-site thermal energy source in the Canadian pulp and paper sector and is also relevant in food processing operations co-located with feedstock supply (Natural Resources Canada, CanmetENERGY 2026). Outside the pulp and paper sector, deployed bioheat in Canada is concentrated at the 50 kW–5 MW thermal capacity range tracked by CanmetENERGY's Canadian Bioheat Database; the 2023 survey update reports that more than 90 % of catalogued commercial, institutional, and small-industrial systems use either wood chips (approximately 45 %, dominant in institutional, agricultural, and light industrial installations) or wood pellets (approximately 46 %, dominant in institutional and commercial applications), with hog fuel and agricultural residues uncommon at this scale and concentrated in pulp and paper and adjacent forest-products operations (Natural Resources Canada, CanmetENERGY 2023). Regional fuel-type preferences are pronounced: wood chips dominate in Quebec, Ontario, British Columbia, and Nova Scotia, while pellets account for nearly all use in New Brunswick and the Northwest Territories, with lesser amounts in Quebec and BC. These distributional facts inform the technologies admitted to the bioenergy supply node feeding the Light Industrial sub-model, and the regional weighting applied to wood-chip versus pellet capital and operating cost assumptions. Sustainable supply is the binding constraint at the system level and is represented in CIMS through energy supply curves rather than at the device level.

34.3.5 Geoexchange and Recovered Waste Heat

Shallow enhanced geoexchange coupled with a heat pump is deployable across all provinces for facility heat. Independent of geoexchange extraction, every industrial site is itself a source of low-grade waste heat — compressors, condensers, flue gases, and effluents — that is conventionally rejected to the environment. National-laboratory analyses estimate that 20–50 % of industrial process heat could in principle be supplied via recovered waste heat (Thekdi and Nimbalkar 2014). Process integration and heat-pump-assisted recovery are represented in CIMS as a service-side reduction in the demand directed to other supply technologies, parameterized by sub-model based on recoverable-heat fractions reported in MECS and CanmetENERGY studies, rather than as a competing supply technology at the service node.

34.3.6 Thermal Energy Storage

Pressurized hot water, molten salt, and solid-media storage shift electrified heat loads to off-peak hours and extend solar thermal coverage to continuous-process duty cycles. Storage is treated in CIMS as a distinct technology that competes at the same service node as the heat-supply technologies it enables, with round-trip efficiency, capacity, and discharge-duration parameters drawn from IEA reference values (International Energy Agency 2022).

34.3.7 Hydrogen

Direct combustion of hydrogen below 200 °C is not competitive with electrification under most modelled conditions: round-trip energy efficiency from electricity to hydrogen to heat is substantially lower than direct electrification or heat-pumps, and the system value of hydrogen is concentrated in genuinely high-temperature or feedstock applications. Hydrogen is therefore not included in the default low-temperature technology set; it remains available in the broader CIMS technology library for the medium- and high-temperature sub-models (Thiel and Stark 2021; Rissman et al. 2020).

34.4 Implications for the CIMS Industrial Sub-Models

The technology competitions in the food and beverage, pulp and paper, chemicals, mining, and light manufacturing sub-models admit, at the relevant service nodes, conventional natuarl gas and (where regionally relevant) heavy fuel oil boilers, used for calibration to CEUD and NIR baseline data; electric resistance and electrode boilers, parameterized by provincial industrial electricity rates; industrial heat pumps disaggregated by sink-temperature class (≤ 100 °C, 100–150 °C, 150–200 °C) with explicit COP and source-temperature assumptions drawn from IEA HPT Annex 58; mechanical vapour recompression where evaporation duties exist; biomass and biogas boilers, with energy availability governed by the bioenergy supply sub-model; and solar thermal in provincial sub-models, parameterized to regional irradiance (Natural Resources Canada 2024; Environment and Climate Change Canada 2024; IEA Heat Pumping Technologies Programme 2023; Natural Resources Canada, CanmetENERGY 2026; Schoeneberger et al. 2020).

Existing boiler stock, distinguished by sub-model and temperature class, is estimated based on CEUD and NIR base-year data. All existing boiler stock is assumed to stay online for modeled lifetimes past the base year, after which replacement is required. Policy or other drivers may drive earlier retirement and replacement of some of these existing units. Capital costs, retrofit feasibility (as defined in the technology chapter), and electricity-price sensitivity vary substantially across these technologies.

35 Cogeneration

35.1 Scope and Definition

Cogeneartion (also referred to as combined heat and power) is the sequential or simultaneous generation of multiple forms of useful energy (usually mechanical and thermal) in a single, integrated system. Cogeneartion systems consist of a number of individual components—prime mover (heat engine), generator, heat recovery, and electrical interconnection— configured into an integrated whole. The type of equipment that drives the overall system (i.e., the prime mover) typically identifies the cogeneartion system. Prime movers for cogeneartion systems include steam turbines, gas turbines (also called combustion turbines), spark ignition engines, diesel engines, microturbines, and fuel cells. These prime movers are capable of burning a variety of fuels, including biomass/biogas, natural gas, or coal to produce shaft power or mechanical energy. Additional technologies are also used in configuring a complete cogeneartion system, including boilers, absorption chillers, desiccants, engine-driven chillers, and gasifiers. Boilers and gasifiers are discussed in Chapter 5 of this document. (U.S. Environmental Protection Agency (EPA) and Partnership (2007))

Although mechanical energy from the prime mover is most often used to drive a generator to produce electricity, it can also be used to drive rotating equipment such as compressors, pumps, and fans. Thermal energy from the system can be used in direct process applications or indirectly to produce steam, hot water, hot air for drying, or chilled water for process cooling.

The industrial sector currently produces both thermal output and electricity from biomass in CHP facilities in the paper, chemical, wood products, and food processing industries. These industries are major users of biomass fuels—utilizing the heat and steam in their processes can improve energy efficiencies by more than 35 percent. In these applications, the typical CHP system configuration consists of a biomass-fired boiler whose steam is used to propel a steam turbine in addition to the extraction of steam or heat for process use.

36 Carbon Capture, Utilization, and Storage

TODO: Describe how the model represents carbon capture, utilization, and storage across sectors, including capture technologies, capture rates, energy penalties, transport and storage infrastructure, and associated costs. The section should explain which emissions sources are eligible for capture, how captured CO_2 flows are tracked, and whether utilization pathways or permanent sequestration are distinguished. It should also document assumptions related to technology availability, deployment constraints, storage capacity, policy incentives, and interactions with hydrogen production, industrial processes, and power generation.

37 Energy Storage

TODO: Describe how the model represents energy storage technologies and their role in balancing supply and demand across the energy system. The section should document the types of storage included (e.g., batteries, pumped hydro, thermal storage, hydrogen storage), storage duration assumptions, charging and discharging efficiencies, capacity constraints, and operational behavior. It should also explain how storage interacts with variable renewable generation, grid reliability requirements, peak demand management, and transmission infrastructure, along with assumptions for costs, learning rates, and deployment limits.

Part V

Data and Calibration

38 Data Inputs

TODO: A reader-oriented description of the `data/` directory. Cover: (1) `data/model_inputs/` — what each region/sector CSV represents, the meaning of the Branch / Type / Parameter / Service / Technology columns, how rows are interpreted as nodes in the CIMS tree, what `service_provide` / `service_request` / `competition` / etc. parameter values mean. (2) `data/mapping_conversions/` — region_primary, sector_map, energy_map, the region aggregation files, and how they're used during processing and rendering (the dynamic sector-by-region table in the architecture chapter is one example). (3) `data/raw_data/` and `data/processed_data/` — source datasets and how they flow into model inputs. (4) `data/calibration/` — what lives there. Cross-reference the file-format appendix for column-by-column specifications.

May want to include information here about including geographic levels other than provinces and territories, like energy markets, other countries (for trade dynamics), and cities (for building and transportation policies).

39 Calibration

TODO: Expand the outline below into a full chapter. Each bullet is a section heading or paragraph topic in its own right — the calibration period and counterfactual data sources, the FIC-optimization workflow, the manual-adjustment fallback, how the forecast-period FIC is derived from the historical estimates, and the long-term FIC trend-to-zero convention all deserve explicit treatment.

- Optimization by year to counterfactual data
- Change FIC values to match objective criteria (counterfactual)
- Can use market share for tech competitions at a node, or energy or emissions at same or adjacent node
- Also have option for manual adjustment of FIC values for more disconnected objective and adjustment locations in the cims tree
- Optimize FIC year by year over historical counterfactual period
- Use median or filtered mean (remove top/bottom 10% before mean) to estimate FIC for forecast period
- Trend all FIC values to zero over very long-term (next 50-70 years?)

Calibration of endogenous supply sectors:

1. Set Financial lifecycle cost to 1
2. Set Price Multiplier to expected price (including sector markup)
3. Calibrate demand sector
4. Calibrate energy supply sector to achieve expected production cost at node one level below top level
5. Remove Financial lifecycle cost
6. Reset Price Multiplier to only include sector markup

40 Validation

TODO: Document how the calibrated model is checked against independent data. Common validation comparisons for CIMS: historical fit against the NRCan Comprehensive Energy Use Database (CEUD), comparison of emissions against the ECCC National Inventory Report (NIR), comparison of new-stock investment against Statistics Canada figures, sensitivity tests for key behavioural parameters (variance v, discount rates). Include a discussion of known limitations and where validation is weak (e.g. provinces with sparse historical data).

Part VI

Running the Model

41 Setup the model

Welcome to the CIMS repository! This guide will help you set up and run the CIMS Python package for energy-economy modelling.

41.1 Prerequisites

Before you begin, ensure you have the following installed:

- **uv**: A Python package and environment manager. Install it with one command — no separate Python installation required.
  - Mac/Linux:

    ```
    curl -LsSf https://astral.sh/uv/install.sh | sh
    ```

  - Windows (PowerShell):

    ```
    powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/i
    ```

- **Git**: Used to clone the repository. Follow the installation instructions for your operating system.
- **Terminal or Command Prompt**:
  - Windows: Git Bash or PowerShell
  - macOS/Linux: Terminal

41.2 Setup

41.2.1 1. Clone the Repository

Open your terminal and navigate to the directory where you want to work:

```
cd ~/Documents/projects/
```

Clone the repository:

```
git clone https://github.com/EMRG-SFU/cims.git
cd cims
```

41.2.2 2. Install Dependencies

From the `cims` directory, run:

```
uv sync --extra notebooks
```

This will automatically select a compatible Python version, create a virtual environment, and install all dependencies. You only need to do this once, or again when dependencies are updated.

41.2.3 3. Launch Marimo

```
uv run marimo edit
```

Marimo will open in your default browser. From there you can open, run, and edit notebooks.

41.3 Common Workflows

I want to...	Command
Set up or update my environment	`uv sync --extra notebooks`
Launch Marimo	`uv run marimo edit`
Run a Python script	`uv run python my_script.py`
Open a Python shell	`uv run python`
Use a specific Python version	`uv sync --python 3.12 --extra notebooks`

41.4 Subsequent Runs

To re-launch Marimo in future sessions, simply run from the `cims` directory:

```
uv run marimo edit
```

uv will use your existing environment and skip reinstalling dependencies unless something has changed.

41.5 Troubleshooting

- **uv not found after install**: Restart your terminal session so the `uv` command is available on your PATH.
- **Dependency conflicts**: Run `uv sync --extra notebooks` again to bring your environment up to date with the latest `pyproject.toml`.

41.6 Additional Resources

- **Submit Issues**: If you encounter any problems, please submit an issue on our GitHub Issues page.

42 Getting started

how to run a scenario

Discuss model operational structure (inputs, marimo notebook, results)

43 Policy Costing & Outputs

43.0.1 Policy costing methods associated with ISTUM and early CIMS

Issues associated with deriving costs from its outputs guided much of CIMS' development. The cost methodologies in the following sections are described in brief and outline some of the issues in estimating costs.

The ISTUM model was initially designed for energy policy analysis, mainly for assessing how fuel consumption and emissions would change with a given energy policy. The emphasis was on marginal costs, not total or absolute expenditures. The first consistent costing concept was the techno-economic cost, or TEC.

Techno-economic Cost (TEC)

TEC is the difference in estimated (ex-ante) financial expenditure on capital, labour, and energy between the Business-as-Usual and Policy cases. It does not usually (unless otherwise specified) contain carbon taxes as these are considered a transfer internal to society. It is possible to get negative TEC costs, or benefits, especially in transportation where increased use of transit and walking and cycling reduces financial costs, implying that some actions induced by policy may be profitable. There are significant complications as well – do those who switch to transit and walking and cycling continue to own and pay for cars? This reflects the challenging question about how vehicle acquisition will be affected by policies to reduce vehicle use. TEC for all consuming sectors are reported with and without electricity price increases, which are sub-regional transfers that balance out to zero at the regional level once the financial effects on electricity producers' revenues are included.

The ISTUM sub-models were integrated under the energy demand and price feedback system in the mid-1999 to 2000 period, thus creating the first version of CIMS. During this period, the model was used for a series of large GHG abatement analyses for Canada. The thrust of these analyses was to integrate and assess a series of emissions reduction actions proposed by panels of interested parties, the Issue Tables of Canada's Climate Change National Assessment Process. The most effective actions were determined by minimizing their welfare costs, as opposed to their pure financial costs. This led to the development of the Perceived Private Cost methodology (Bataille et al. 2002, Laurin et al. 2003, MKJA/EMRG 2000).

Perceived Private Costs (PPC)

PPC are calculated as the GHG tax multiplied by the marginal amount of GHG reductions that occurred at each GHG tax level. The area under the resulting GHG abatement cost curve represents the perceived cost of the actions by the GHG charge. The PPC methodology estimates welfare losses that represent in part the unwillingness of all consumers to switch to technologies that reduce emissions. This fostered development of the Expected Resource Cost methodology, which attempts to more realistically depict the costs and risks associated with a given GHG reduction policy.

Expected Resource Costs (ERC)

ERC is PPC that has been adjusted for risk, option value, and general inefficiency. It is equal to TEC+(PPC-TEC)*0.75. The difference between TEC and PPC that is allowed as ERC is a "rule-of-thumb" estimate for what percentage of the difference can be considered real equipment failure risk, as opposed to inefficient resistance to price signals. It represents that perceived cost, which could turn into real financial cost. Negative ERC is possible, especially at lower GHG charges, but not likely (Bataille et al. 2002, Laurin et al, 2003, MKJA/EMRG, 2000).

Table 10 summarizes the three early costing methods used for CIMS.

Table 8. Types of Costs Used in Early Incarnation of CIMS

Distribution of Costs Under Electricity and Refining Demand Changes

Electricity demand changes in all sectors in response to increased electricity prices, leading to an issue of how to distribute GHG reductions as well as TEC and PPC charges. GHG reductions from reduced electricity consumption were accorded to the direct consumer (based on the policy GHG intensity of electricity), who faced increased electricity prices, while electricity producers were accorded the remaining GHG intensity reductions, the TEC costs generated by their intensity reductions plus their changes in revenue. These net transfers, while positive at the sub-regional level, disappear at the regional level.

The Relationship Between TEC, ERC, Option Value and Welfare Costs

The cost difference between ERC and TEC estimated during early analyses was described as risk, or "failure risk", which is related to the concept of option-value (Pindyck 1991, Dixit 1992, Dixit and Pindyck 1994). Option value is the expected gain from delaying or avoiding an investment while waiting for new information that might lead to a better decision. Thus, ERC are anticipated ex-post costs, and it is calculated as the sum of expected financial costs (TEC), lost option value, consumers' surplus losses internal to residential and transportation (internal shifts away from business-as-usual (BAU) preferences within the sub-models) and finally, when demand feedbacks are included, consumers' surplus losses related to demand adjustments for domestic consumers of domestic goods. It may therefore be considered a measure of personal welfare costs, not including negative environmental externalities (e.g. poorer quality air) or positive public goods (e.g. education, defence or public parks).

43.0.2 Policy Costs Associated With Changes in Demand

The inclusion of final demand feedbacks required the addition of five new policy-costing concepts. In all these concepts, financial output is defined as a sector's sales revenue, which includes both value-added and purchased inputs. Value-added is net profits plus the cost of all expenditures on land (unprocessed "gifts of nature"), labour and capital (Lipsey et al. 1988).

- The cost of production effect on sector financial output, which is the increased financial expenditures required to adapt to the policy, or TEC.
- The demand effect on sector financial output, which is the reduction in production expenditures associated with lower output.
- The absolute change in sector financial output, which is the sum of the cost of production and demand effects.
- Consumers' surplus losses associated with demand changes in transportation and residences.
- Changes in value-added associated with cost of production and demand changes in manufacturing.

43.0.2.1 The Cost of Production and Demand Effects on Sector Financial Output

The cost of production effect captures the internal capital investment and fuel choice adjustments of the model, or the TEC costs endogenous to the model. It is the anticipated financial cost of technical adjustment to the policy. The demand effect captures the effects of the demand feedbacks system, or reduction in output, as measured by expenditure compared to the business-as-usual case (BAU). The third is the sum of the first two. These three concepts capture a portion of the anticipated financial cost of a policy. They do not include the cost of buying emission permits from foreign sources or the cost of imports that replace domestically produced goods for domestic consumption. The fourth item, consumers' surplus losses, represents lost consumer welfare associated with reduced residential expenditure and mobility, while the fifth represents the lost value-added to manufacturing firms due to policy.

Most GHG policies will increase the cost of production for a given service, increase its price, and thus reduce its demand. This increase in the cost of production will also include the value of carbon charges that are imposed on GHG emissions. These charges are transfers and as such must be subtracted from regional and national costs. Additionally, if demand is inelastic (unresponsive to price) for a given service, the dollar value of the sector, as valued by cost of production, may increase faster than demand reductions, leading to an overall increase in sector value to the economy. If demand is elastic, the opposite may occur.

The standard outputs from the sub-models include BAU and policy values for new physical capital investment associated with consuming energy, labour associated with the latter capital, and energy costs, as well as changes in physical output. It also produces BAU and policy costs per unit output, including GHG charges. These GHG charges are isolated by calculating policy emissions multiplied by the GHG price. When the micro and macro economic dynamics are turned on, part of the difference between the BAU and policy cases may be physical output changes while the rest will be cost of production differences due to internal stock adjustments. CIMS only captures a portion of the price that the consumer sees; both the BAU and Policy cost per unit are adjusted accordingly. The cost of production and demand effects on sector financial output (SFO) (Equations 23 & 24), and the absolute change in financial expenditure are calculated as follows (Equation 25).

(Equation 28)

(Equation 29)

(Equation 30)

Where:

QGHG = GHG emissions (Tonnes CO2e)

PGHG = GHG price ($/Tonne CO2e)

Qpolicy = Policy physical output of good or service X

QBAU = BAU physical output of good or service X

Ppolicy = Policy cost per unit of good or service X

PBAU = BAU cost per unit of good or service X

The absolute change in sector financial output could be positive or negative depending on the elasticity of demand – if the cost of production rises faster than demand falls, then the absolute change in financial output could be positive. This result is common with less aggressive policies. The demand effect on sector financial output will always be negative for normal goods, whose demand fall as their price rises. The cost of production effect on sector financial output is usually positive, except for the case where there are decreasing returns to scale.

43.0.2.2 Consumers' Surplus Adjustments in Residential and Transportation

Reductions in residential activity and mobility reflect consumers' surplus losses. Consumers' surplus is the area between the demand curve, which symbolizes the marginal willingness to pay for a service or good, and the market-clearing price of the good or service (Figure 43.1).

The consumers' surplus loss between BAU and Policy is then calculated as BAU minus the policy. When calculating the direct welfare costs of a policy, consumers' surplus loss is added to the ERC value already calculated. Table 11 summarizes the financial and welfare costs adjustments made to the various sectors when their output changes.

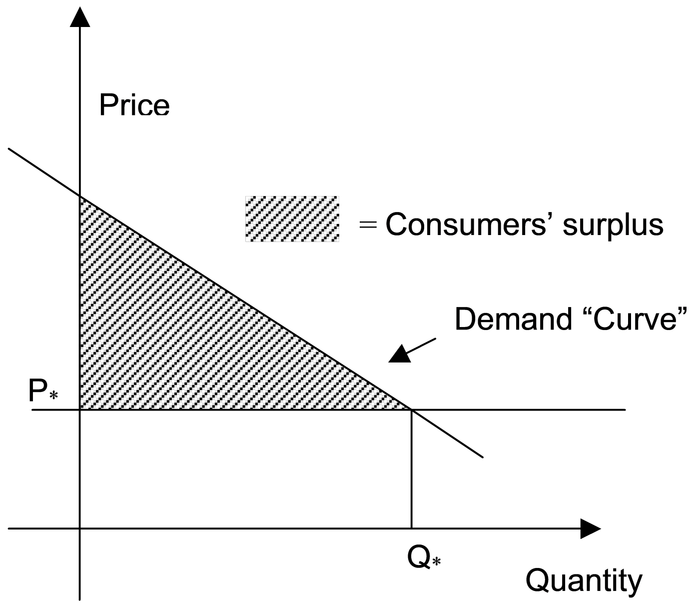

Figure 43.1: Consumer Surplus

Table 43.1: Financial, welfare cost and value-added adjustments induced by changes in output

	Output adjustment to financial costs	Output adjustment to welfare costs / value-added
Personal Transportation	Demand effect on sector financial value, as measured by expenditure	Difference in **consumers' surplus** under BAU and policy calculated using transportation *pkt* elasticity
Industry (& Freight Transportation)	Demand effect on sector financial output, as measured by expenditure	Difference in **value-added** between BAU and Policy
Residential	Demand effect on sector financial value, as measured by expenditure	Difference in **consumers' surplus** under BAU and policy calculated using combination of savings/consumption and leisure/work elasticities
Commercial	Demand effect on sector financial value, as measured by expenditure	None

The lack of an output adjustment for welfare costs or value-added for the commercial sector is partly due to the nature of the Commercial and Institutional sub-model. It is composed mainly of building shells, HVAC, miscellaneous electricity and air-conditioning systems for government and office buildings, retail space, hospitals, schools, etc. CIMS does not model what these spaces are used for, and thus cannot account for how financial output or value-added may have changed.

43.0.3 Outputs from CIMS

Table 12 provides the outputs specific to the sub-models (i.e., for when they are run by themselves without feedbacks), while Table 13 provides the extra outputs necessary for working with the CIMS' micro and macro economic systems of dynamics (also referred to as feedback systems).

Table 10. Sub-model Outputs Existing Prior to This Research

Sub-models Annualized lifecycle costs by sector by year by technology based on behavioural discount rates (20-80%) (See Table 5).

Fuel use and emissions by sector by year by technology.

Service and auxiliary use by sector by year by technology.

Total stock use by sector by year by technology.

New stock use by sector by year by technology.

Retrofits by sector by year by technology.

Technology cost changes due to declining capital cost function

Table 11. Supply and Demand Feedback Outputs Added as a Result of This Research

Supply and Demand feedbacks Annualized lifecycle cost per unit produced for all sectors by year based on financial cost of capital for both the BAU and Policy case; the model operator sets the cost of capital (e.g. 7.5 or 10%)

BAU and Policy final demand for each sector's product or service by year.

BAU and Policy disaggregated energy demand for all sectors by year.

Changes in disaggregated capital, operations and maintenance, and energy costs by sector by year.

Changes in GHGs and selected criteria air contaminants (CACs), energy and service use by technology, sorted by technology competition, for BAU and Policy by sector by year.

Changes in net exports of key energy commodities from BAU to Policy

Changes in energy prices by year

Changes in the financial cost of production split into the cost of production and demand effects for industry and manufacturing

Changes consumers’ surplus for personal transportation and residences

Changes in value-added for industry and manufacturing

44 Building the Model Tree

TODO: Write up best-practices for designing nodes in the CIMS tree. Cover: when to request services directly at a node vs. through technology competition, how to lay out service nodes for a new sector, naming and Branch-path conventions, where intangible costs are typically anchored, and how to expose policy levers cleanly. Worked examples (one residential heating, one industrial process-heat) would communicate the patterns better than abstract prose.

> **Note**
>
> If setting market shares after the base year, need to use lifetime = 1, otherwise stock is not calculated correctly.

Part VII

Outputs and Analysis

45 Outputs

Three main classes of output are created by the model:

1) Sub-sector specific reports

   a) fuel and emissions for a given sector (FUELA) for BAU and Policy, (B or C-Region-FUELA, e.g. BONFUELA.

   b) technology levelized costs for a given sector (LEVEL) – same pattern as FUELA.

   c) new and total technology market share reports for a given sector (NEWMA and TOTAL) – same pattern as FUELA.

   d) technology node competition reports for a given sector (REGION-NC-YEAR, e.g. "ABNC2005"

2) Semi-aggregate reports

   a) changes in physical output, sorted by sector and by region (CCOMP)

   b) changes in sector capital, fuel, emissions and labour costs, sorted by sector and by region (CIMCOSTA & CIMCOSTB)

   c) changes in sector energy use (by market class Elec., NG, coal and RPP) and GHG emissions, sorted by sector and by region (DCOMP)

   d) changes in sector cost per unit (inclusive of emissions charges), sorted by sector and by region (LCOMP)

3) Aggregate report

   a) CIMS Outputs (As of March 2007)

45.1 Sub-sector Reports

Four report types are available. In each type of report, the analyst opens a spreadsheet where data can be manipulated using spreadsheet functions. The report can then be reformatted, printed, or saved under another name.

45.1.1 Fuel and Emissions Report

Data concerning fuel consumption and emission generation are saved for each year of the simulation. The spreadsheet entitled B(BAU) or C(Policy) Region (e.g. BC) FUEL, e.g. CBC-FUELA, receives these data. The model's report generator produces a table containing: (1) a list of all fuels and emissions, (3) consumption/production of those fuels and emissions for each year, and (4) a summation of fuels consumed and emissions output. Note that, if fuel/emission units are not uniform, this summation will be erroneous. Figure 3.1 displays the layout of a typical aggregate fuel report.

Figure 3.1 ISTUM's Fuel Report

Typically, technologies are grouped into Aggregate, Major processes, Service categories and Product type categories.

Aggregate reports include the cumulative energy consumption of all the technologies in the industrial branch under analysis and are generated for every industry. Note that this is the sum of consumed and produced energy. To determine the amount of cogenerated or self-generated energy, the analyst should refer to the NC files which show al energy inputs and outputs by technology competition. No technology can contribute to the cumulative consumption/production of more than one group in each category.

Figure 3.1 ISTUM's Fuel Report

45.1.2 Total Stock Report

The table generated in the TOTAL spreadsheets contains data on the sum of operating old and new stock inventories for that year. The stock is listed in output units for that technology. As with the other report types, the table's header contains specific information on the scenario and economic criteria applied to produce that output. To the right of the total stock table, one finds a table of the share of total stock for competing technologies (in percent) that each technology has obtained. Figure 3.23 displays the layout of a typical Total Stock Report.

The lower portion of this table provides information concerning changes in stock consequent to retrofitting. Each set of lead and retrofit technologies show the lead technology first followed by the technology(ies) that can be retrofit to it. The data indicate the number of units not retrofit (the value associated with parent technologies) and those that were retrofit (the values associated with the retrofit technologies).

Figure 3.2 ISTUM's Total Stock Report

45.1.3 New Stock Report

As in the Total Stock Report, the two tables generated by the New Stock Report function of CIMS, show new stock additions in each year of the simulation and the market share of that stock purchased in that simulation year. This NEWMAR spreadsheet header, like the others, displays specific information concerning scenario and economic criteria. Figure 3.3 displays the layout of a typical New Stock Report.

Figure 3.3 ISTUM's New Stock Report

45.1.4 Levelized Costs Report

The LEVEL report lists each technology and contains tables displaying two categories of information:

1) The first set of columns include data used to generate the levelized costs. The first numeric column displays capital costs (CC) of each technology, followed by its annualized CC (CC x CC recovery factor, found in the sixth column). The third column shows the sum of annualized CC and costs of operations and maintenance (excluding fuel and emissions costs) for each technology. This, divided by the material output found in column 4 generates the lifecycle cost per unit output in column 5. Column six indicates whether any non-cost parameter was applied to the capital costs (CC x non-cost parameter).

2) The second set of columns, each headed by a simulation year, presents the levelized cost of that technology in that year. The levelized cost is the sum of the per-unit cost (column four) and the annual fuel (and emission) costs. (Figure 3.4)

The lower portion of this table provides information concerning retrofits applied in that simulation. Each set of lead and retrofit technologies, displayed in separate groups, show the lead technology first followed by the technology(ies) that can be retrofit to it. Note that the capital cost of the lead technology is always 0 since it is considered a sunk cost. The capital costs associated with the retrofit technologies represents the cost of retrofit purchase and installation.

Figure 3.4 ISTUM's Levelized Cost Report

45.1.5 Node Competition Report

There is a separate node competition report for every sector in every region in every year. They are title (region)NC(year)J, e.g. for all sectors in Alberta the 2005 report is called "ABNC2005J.xls". The region codes are: BC, AB, SK, MB, ON, PQ, ATL (for Atlantic).

The node competition report presents the results of a run as a comparison between BAU and POLICY in terms of fuel use, services and emissions. The key difference between this report and the last four is that it organized by technology competition. As there a several fuels, services and emissions in CIMS, these reports tend to be very large. Figure 6 on the next page provides a sample node competition report for the baseload electricity production competition in Alberta in 2005. The shown figure gives the difference in stock, fuel use and emissions, while further to the right on the spreadsheet will be the BAU and policy absolute values. Stock is valued in the output unit of the sector in question, energy in GJ, and emissions in tonnes.

Figure 6: Sample node competition report – base load electricity production Alberta 2005

45.2 End user reports

Reports of energy use, financials, intensities, travel, etc. at end user level

For example, household energy spending, average travel and costs

MACC Curve like in EDF report, MACC 2.0

45.3 LMDI Decomposition

Can this be done appropriately at the sector level or only at region/national?

Decomposition of results (to also use in policy attribution)

Include automatic waterfall chart?

Factors:

- Activity
- Etot / Activity (efficiency)
- Eff / Etot (fuel switch)
- GHGtot / Eff (fuel decarbonisation)
- GHGnet / GHGtot (CCS?)

45.4 Attributing emissions reductions to policies

Shapley values using SHAP?

46 Post-Processing and Reporting

TODO: Describe the post-processing workflow that turns raw CIMS outputs into the figures and tables typically presented in a study. Likely includes: pivoting and aggregating output CSVs, computing derived metrics (emissions intensities, levelized costs, sector totals), comparing scenarios against a baseline, building standard chart sets (energy demand by sector, technology stock evolution, emissions trajectories). If any scripts in the repo support this, link to them and show example usage.

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A Acronyms

TODO: Single source-of-truth list of acronyms used throughout the book. Build this by walking the existing chapters and pulling out abbreviations on first use — CIMS, ISTUM, LCC, FIC, DIC, DCC, GCC, AEEI, CRF, BS, NS, MS, NMS, PR, CEUD, NIR, NRCan, ECCC, IEA, MVR (mechanical vapour recompression), COP (coefficient of performance), and so on. Use a Pandoc definition list (the same `.eq-vars`-style markup used for equation variables) so the list renders as a clean two-column glossary.

B File Format Reference

TODO: Column-by-column specification of every file format the model reads or writes. At minimum: model_inputs CSV (the 24-column Branch/Type/Region/Sector/Service/Technology/Parameter/Co structure), region_primary and region_aggregation_* mapping CSVs, sector_map CSV, energy_map and energy_conversions CSVs, nir_to_cims_map CSV. Note the "navigate by typing" banner on the first line of every model_inputs CSV (which is why all read code uses `skiprows=1`). Document the expected datatypes and any required vs. optional columns.

C Equation Summary

TODO: A quick-reference table listing every numbered equation in the methodology chapters with its label, name, and one-line description. Sortable by label (`eq-stock_retirement_existing`, `eq-market_share_new`, etc.). Lets readers locate an equation by name without scanning the methodology chapters. Could also be auto-generated from the qmd source by parsing `$$...$$ {#eq-...}` blocks.

D Version History

TODO: A chronologically-organized log of substantive changes to the model and the documentation. Start with the current "version" and seed it with the recent reorganization (per-chapter bibliographies, equation reformatting, dynamic region-sector table, etc.). When a new model release is cut, add a section documenting what's new, what's deprecated, and any migration steps needed for inputs or scenarios from previous versions.